CHESTER
UNIVERAL MILLING MACHINE

VICTORY 360

Operation Manual

Chester UK Ltd
Clwyd Close, Hawarden Industrial Pk
Hawarden, Nr Chester
Flintshire. CH5 3PZ
Tel: 01244 531631
Email: sales@chestermachinetools.com
www.chestermachinetools.com

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1. Safety Instruction

1.1 General safety rules

Operator must read the instruction carefully before operating the machine, and the manager of safety department should assure the operator knows the requirements well.

WARNING: Never contact the cutters and other driving parts by hands if they are still moving.

1.1.1 The operation, maintenance and repair of the machine must be done by qualified person who have been trained to have the ability to forecast the potential risks. Only safety-conscious persons who are fully aware of the risks can operate the machine.

1.1.2 To those people who assemble, operate, or maintain the machine must be confirmed to have read and understood the operating instructions.

1.1.3 After stopping the machine, the tool will still run for a period of time due to inertia, do not open the tool guard and touch the tool by hands before it comes to a stop.

1.1.4 Do not detach or modify any part of the safety guard. The machine should be disconnected from the automatic supply during maintenance or repair.

1.1.5 Assign special persons to operate, maintain and adjust the machine.

1.1.6 Operate the machine in technically perfect condition. Only relative professional can completely repair the machine.

1.1.7 Stop the machine immediately if abnormal phenomena appears, check and repair the machine by professional in time.

1.1.8 Convey the machine with the hoisting equipment, which is of enough load capacity.

1.1.9 Observe all safety instructions and warnings attached to the machine, make sure they are always complete and perfectly legible.

1.1.10 Do safety inspection before operating the machine. Ensure the stroke limit

switches, stop block and emergency stop button is safe and reliable.

1.1.11 Reinstall the removed guards and safety devices in place after maintaining the machine.

1.1.12 Do maintenance or adjustment only after shutting off power source.

1.1.13 Always keep children away from the machine. Operator must be at least 18 years old.

1.1.14 Do not wear loose clothing, gloves, neckties or jewelry (rings, watches, etc). Keep the sleeves and the edges of the work uniform tight. Always be sure to wear safety goggles and wear safety shoes during operation.

1.1.15 Put the long hair into a safety cap during operation no matter the operator is a man or a woman.

1.1.16 It is recommended to wear suitable hearing protection equipment if necessary.

1.1.17 Provide adequate light around the machine, and keep the perimeter around the machine dry, clean and in good order. In addition, do not place anything near the machine; otherwise it becomes an obstacle during operation.

1.1.18 Do not remove the guards from their working place during operation.

1.1.19 Disconnect the automatic supply before leaving the machine.

1.1.20 Restart the machine only after the guards and safety devices are replaced to their position and in their good function.

1.1.21 Do not place tools, work pieces, or other items, which is not in use on the machine, especially on the moving parts.

1.1.22 Clamp workpiece and tools completely and reliably and take off the wrench from chuck before starting the spindle running.

1.1.23 Stop the machine before adjusting the position of the coolant nozzles.

1.1.24 Do not use compressed air to remove dust and chips, deposit on the machine, switchboard, control unit etc.

1.1.25 Both operators and the maintenance men should read the descriptions carefully on the caution plate attached to the machine. They should observe the OPERATION MANUAL during their work. During operation and maintenance,

care should be taken so that the caution plate is not dirtied or damaged.

1.1.26 Always bear in mind where the emergency stop button is so that you can operate it without delay in case of an emergency.

1.1.27 Start the machine according to the starting procedures strictly.

1.1.28 Keep your hands away from the turning parts of the machine during operation.

1.1.29 Do not remove the chips directly by hand, It is safer with proper tool to remove the chips. Be sure to stop the machine when removing chips from the machine.

1.1.30 Prior to starting test operation, read this manual carefully so as to be familiar with this machine.

1.1.31 Contact the manufacturer if for some reason the OPERATION MANUAL should become unreadable.

1.2 Additional Safety Rules for Mills

1.2.1 Read and understand the entire instruction manual before operating the machine.

Warning: Failure to follow the instruction will result in serious injury.

1.2.2 Always wear approved safety glasses/suit while operating machine tool.

1.2.3 The machine must be grounded.

1.2.4 Before operating the machine, remove tie, rings, watches, other jewelry, Keep the sleeves and the edges of the work uniform tight. Always be sure to wear safety goggles and wear safety shoes during operation. Do not wear gloves while operating machine tool.

1.2.5 Keep the floor around the machine clean and away from scrap materials, oil and grease, etc.

1.2.6 Keep all guards of the machine securely in place at all times when working. For maintenance purpose, use extreme caution to replace the guards immediately after maintenance.

1.2.7 Make sure workpiece and cutter be fitted and clamped well, and make sure the cutter doesn't touch the workpiece before starting up the machine.

1.2.8 Must shut off power supply before adjusting or maintaining the machine.

1.2.9 The operator must keep clear-minded when operating the machine and pay

attention to what he is doing. It is not allowed to operate the machine when the operator is tired, after drinking or taking medicines.

1.2.10 Use tools properly. Don't force a tool or an attachment to do work which was not designed. Sharp tools should be used. Deformed or dull tools should not be used.

1.2.11 Make certain the motor switch is in OFF position before connecting the machine with power supply.

1.2.12 Never attempt to operate or adjust the machine when the procedure is not understood.

1.2.13 Always stop the machine before adjusting the position of the cooling nozzle.

1.2.14 Operator should change machining position on the table frequently in order to prolong the life of the machine.

1.2.15 Lubricate the machine in time according to the lubricating demand.

1.2.16 Keep the electrical elements clean, do not clean the electrical elements with kerosene or gas.

1.2.17 Be sure the spindle rotation of the machine comes to rest before changing spindle speed.

1.2.18 It is forbidden to process flammable and explosive metal, for example: pure aluminum, magnesium and so on.

1.2.19 The machine should never be used in flammable, explosive or humid environment.

1.2.20 Make sure the operating site should keep well ventilated. It is recommended that ventilation equipment should be provided on the operating site.

2. Working Environment of Machine

2.1 The milling machine is designed for operating on the site:

—The height above sea level doesn’t exceed 1000m.

—The ambient temperature range doesn’t exceed 5℃～40℃.

—The relative humidity doesn’t exceed 50% within＋40℃ and 90% within＋20℃.

—Transportation temperature range is within -25℃～+55℃

__ The light in the working environment should not be lower than 500 lux.

2.2 Do not use the machine in an environment of electric dirt, explosion, metal fretted, gas and steam which can destroy insulation.

2.3 Do not use the machine in an environment of impact and vibration.

3. Purpose and Structure of Machine

The universal swivel head milling machine is designed for milling on general metal workpiece.

Warning: Do not process flammable and explosive metal, e.g. pure aluminum and magnesium, etc.

The universal swivel head milling machine consists of universal swivel head, base, column, knee, table, revolving bed, main drive structure, feed box, ram, coolant, lubrication, electrical system and so on.

The structure of the machine: (Figure.1)

3.1 The column is fixed on the base with screws.

3.2 Knee is in front of the column, and connects with the column through rectangular guideways, which can rise and lower along vertical guideway.

3.3 Revolving bed connects with knee through rectangular guideways. Table connects with slide through dovetail guideways. Worktable and the slide can be moved through lead screw and nut.

3.4 Main transmission adopts gears drive structure.

3.5 The feed box is fixed at the bottom & right of revolving bed, and it is driven directly by motor,

3.6 The coolant system consists of coolant pump, cooling tube and tank in the base and so on.

3.7 Lubrication system consists of oil immersed splash lubrication, lubrication pump and hand pump and so on.

3.8 Operation panel is on the left of the column, which enables convenient operation.

The table is moved by hands or power, the power of feed box is provided by only one motor. The feed box has twenty-four kinds of speed steps and three kinds of rapid speed. Main transmission adopts gears structure, high efficiency, big torque, enlarges speed change range, so enlarge the machining range.

3.9 Rotation of table: Loosen the four screws at two sides of the revolving bed, then turn the table forcibly in target position.

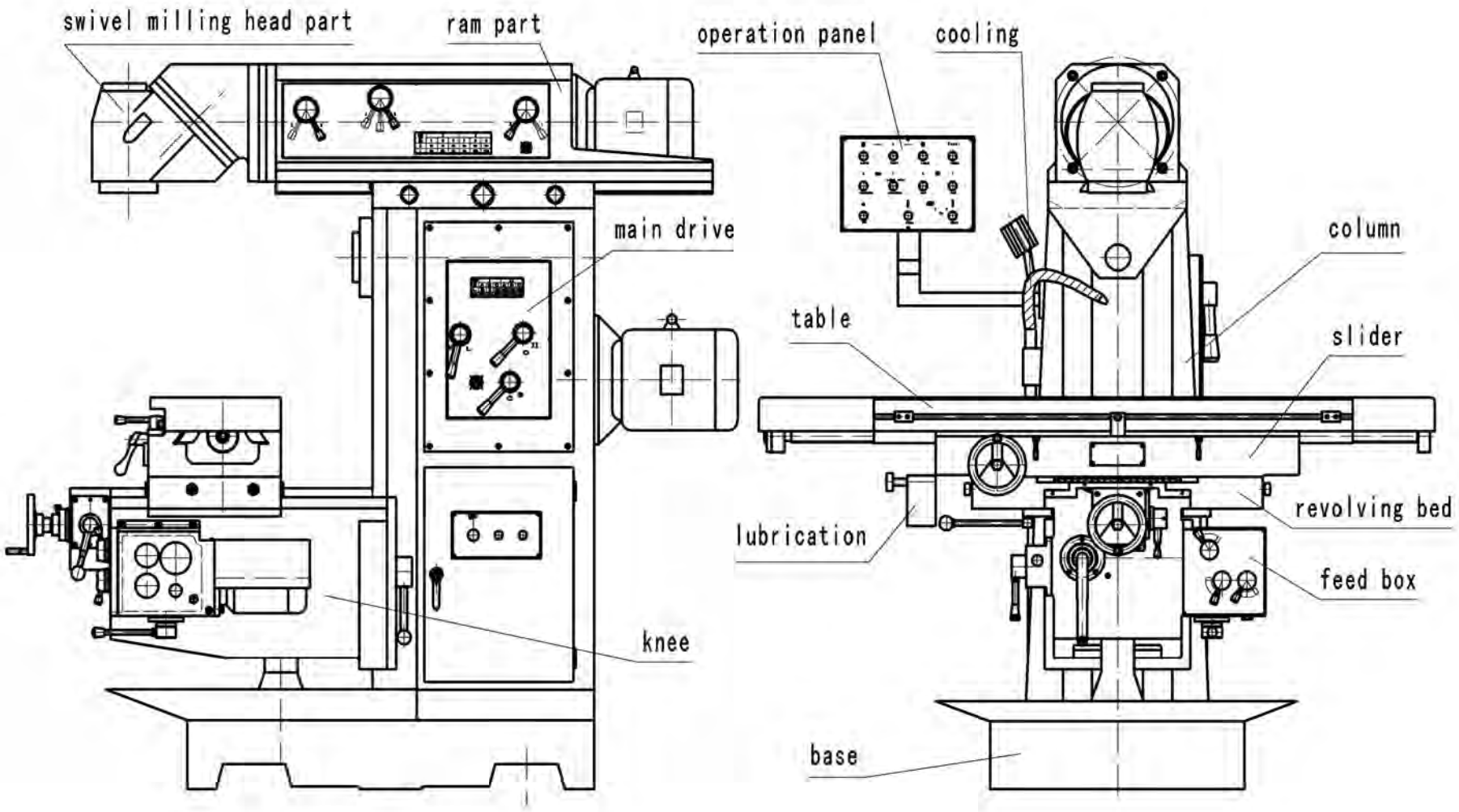

swivel milling head part
ram part
operation panel
cooling
main drive
column
table
slider
revolving bed
lubrication
feed box
knee
base

Figure.1

4. Main Parameter of Machine

No.	Specification	Parameter
1	Spindle taper	ISO50
2	Distance between spindle axis and table surface.	0～400mm (0 ~ 15 - 3/4")
3	Distance between spindle axis and surface of ram	175mm (6 - 7/8")
4	Spindle speed	58 ~ 1800 r.p.m (50Hz);70 ~ 2160 r.p.m (60Hz) 60~1750 r.p.m (50Hz) (universal swivel head) 72~2100 r.p.m (60Hz) (universal swivel head)
5	Table size	1320mm×360mm (51 - 31/32"×14 - 3/16")
6	Table travel (X,Y, Z)	1000mm (39 - 3/8") (longitudinal) 300mm (11-13/16") (cross) 400mm (17 - 3/4") (vertical)
7	Rotary angle of table	±35°
8	Longitudinal, cross, vertical table power feed speed	22 ~ 420 mm/min (7/8" ~ 16-1/2")(50Hz) (X) 22 ~ 393 mm/min (7/8"~ 15-1/2")(50Hz) (Y) 10 ~ 168 mm/min (3/8"~ 6-5/8")(50Hz) (Y) rapid feed 1290mm/min (50-3/4") (50Hz) (X) rapid feed 1205mm/min (47-1/2") (50Hz) (Y) rapid feed 513mm/min (20-1/8") (50Hz) (Z) 26 ~ 505 mm/min (1" ~ 19-7/8")(60Hz) (X) 26 ~ 470 mm/min (1"~ 18-1/2")(60Hz) (Y) 12 ~ 200 mm/min (1/2"~ 7-7/8")(60Hz) (Y) rapid feed 1548mm/min (60-15/16") (60Hz) (X) rapid feed 1446mm/min (57") (60Hz) (Y) rapid feed 615mm/min (24-1/4") (60Hz) (Z)
9	Ram travel	500mm (19 - 5/8")
10	Main motor power	4 kW

11	Overall dimension(L×W×H)	1900 mm×1720 mm×1950 74 - 7/8"×67 - 3/4"×76- 3/4"
12	Net weight	2200 kg
The specifications are subject to modification and improvement without notice.		

5. Transmission System of Machine (Figure.2)

5.1 Universal swivel head

Spindle of universal swivel head drive system is installed in the ram and driven by a flange motor.

5.2 Spindle transmission system

Spindle drive system is installed in the column and driven by a flange motor. The motor connects with a drive shaft and passes power to the spindle through gears and sliding gears.

Main transmission: main motor→drive shaft→gears and sliding gears→spindle.

5.2 Table feed transmission system

Power is produced by manual operation or automatic operation, Power is passed to lead screw and nut to realize longitudinal and cross feed of table.

5.3 Knee

Knee can rise and fall vertically through two bevel gears. Power is passed to lead screw and nut to realize vertical feed of table.

5.4 Ram

Slack the locking handle on the right of ram, turn gear shaft on the right of the ram to adjust ram in the desired position, tighten the locking handle finally.

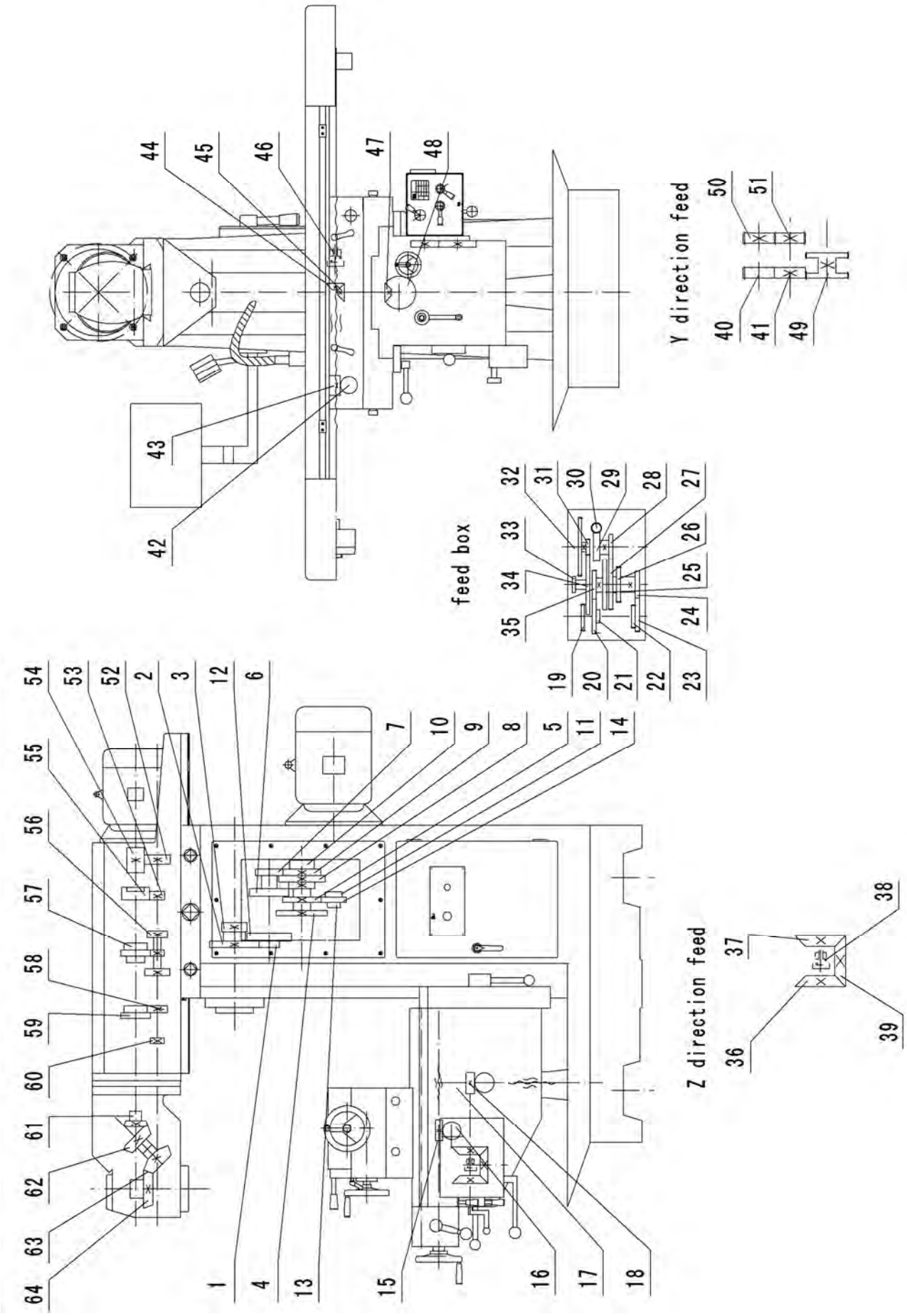


Figure.2

6. Operation of Machine

6.1 Please read the manual carefully before operating the machine, then make familiar with structure of the machine, function of each hand lever, and buttons on the operation panel, coolant & lubricating system and electrical system and so on. (Figure.3)

6.2 Check whether each locking structure is normal before starting the machine, and whether ground wire is right and reliable.

6.3 Switch on power switch, check whether electrical switch is flexible and reliable. Operation panel situates on the left of the column. The pushbuttons on the operation panel have visual symbols, so operation is convenient. There is inching button on the right side of the column, so change spindle speed well.

6.4 Speed change of spindle.

Press inching turning “T” button on the right side of the column first, then adjust three handles (19) in target position according to the illustration of the speed plate.

Spindle speed change of universal swivel head: Adjust three handles (7) in target position according to the illustration of the speed plate.

6.5 Start motor of the feed box first, then adjust handle (1) in “STOP” position, adjust handles (2) at target speed. Table will rapidly feed when handle (1) is situated in “RAPID” position.

6.6 Longitudinal feed of table

6.6.1 Longitudinal power feed of table: First loosen locking handle (4), and adjust the handle (5) in LEFT or RIGHT, NEUTRAL position, which will realize the longitudinal power feed.

6.6.2 Longitudinal manual feed of table: First loosen locking handle (4), and adjust the handle (5) in MANUAL position, then turn handwheel (13) directly to realize longitudinal manual feed of table.

6.7 Cross feed of table

6.7.1 Cross manual feed of table: First loosen locking handle (16), adjust the handle (3) in “NEUTRAL” position, then turn handwheel (21) directly to realize cross manual feed of table.

6.7.2 Cross power feed of table: First loosen locking handle (16), must draw out the knob at the end of handle (3) before adjusting the handle (3) up or down, which will realize the cross power feed.

6.8 Vertical feed of table

6.8.1 Vertical manual feed of table: First loosen locking handle (12), adjust the handle (17) in “NEUTRAL” position, then turn handwheel (18) directly to realize vertical manual feed of table.

6.8.2 Vertical power feed of table: First loosen locking handle (12), adjust the handle (17) up or down, which will realize the vertical power feed.

WARNING: Must take off the crank handle (18) when crank handle (18) is not used.

6.9 Swivel table

First loosen the locking screw (20) at two side of the revolving bed, then turn the table forcibly in target position, tighten the locking screw (20) finally.

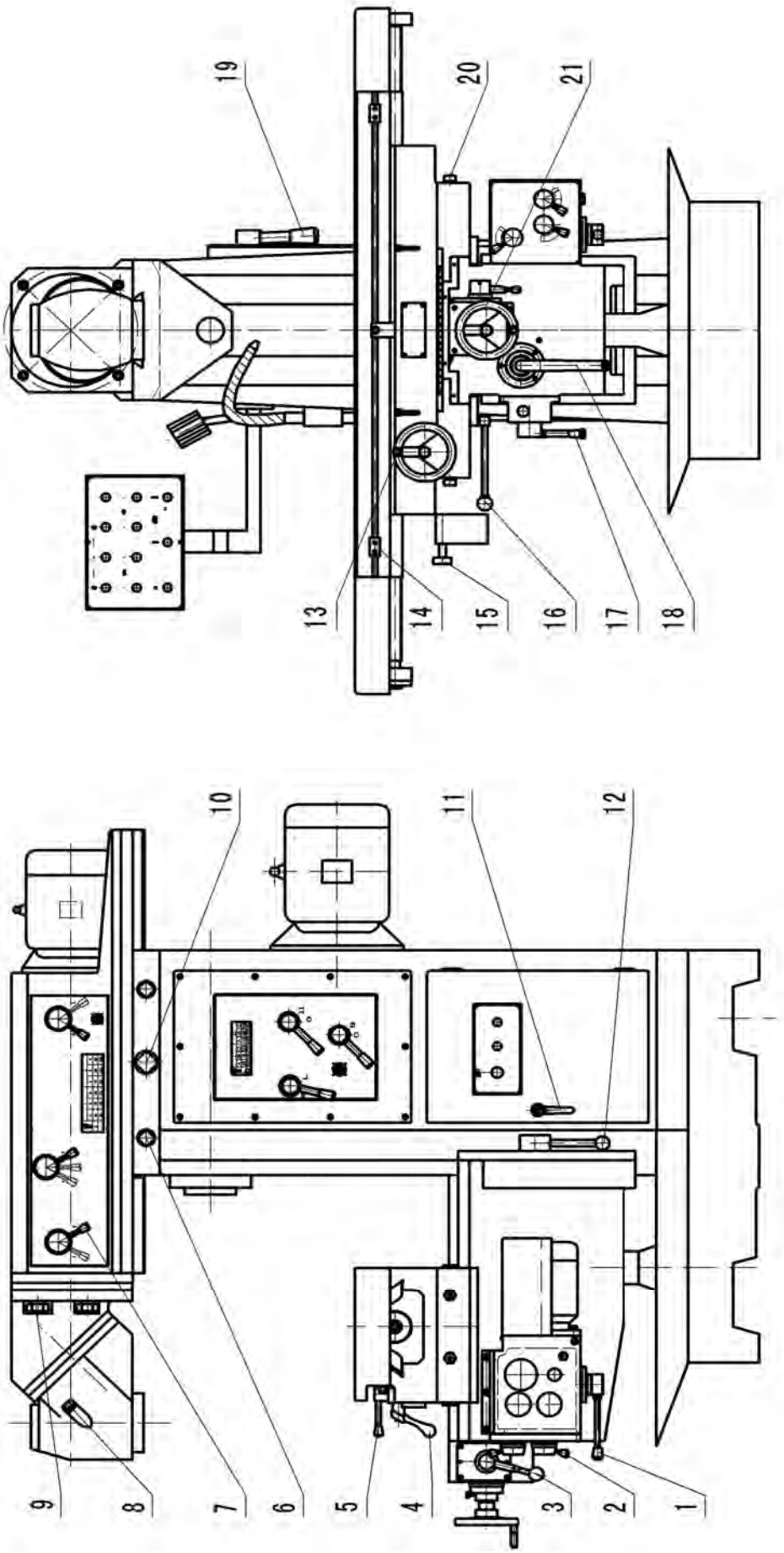


Figure.3

7. Adjustment of Machine

7.1 Adjustment of backlash of the spindle bearings. (Figure.4)

Backlash of spindle bearings has been adjusted well before the machine left the factory. Long time use of machine causes natural wear of spindle bearings and backlash will increase. Please ask professional to adjust it.

Remove cover ① on the right side of the column first. First make claws of lock washer flat under the round nut, adjust round nut (2) to make backlash of spindle proper, then make claws of lock washer turnup, then fix cover (1).

7.2 Adjustment of longitudinal and cross gibs of the table

7.2.1 Longitudinal gib adjustment: First loosen screw ① of small end of gib, then adjust screw ② of big end of gib in proper position until a slight drag is felt while moving the table, tighten screw ① properly finally. (Figure.5.a).

7.2.2 Cross gib adjustment: Remove scrap iron wiper ②, loosen screw ① on one end of gib, adjust screw ③ on the other end of gib to proper position until a slight drag is felt while moving the table, then fix scrap iron wiper ②. (Figure.5.b)

7.2.3 Adjust the clearance between the knee and vertical guideway.

Remove scrap iron wiper ③, loosen screw ① on small end of gib, then adjust the screw ② on the big end of gib in proper position until a slight drag is felt while moving the table, fix scrap iron wiper ③ finally. (Figure.5.c)

7.2.4 Adjustment of the backlash between the longitudinal direction lead screw and nut.(Figure.6)

Too big clearance between the lead screw and nut will affect machining accuracy.

Loosen locking screw first, adjust adjusting worm properly, tighten locking screw.

7.2.5 Adjustment of the backlash between the cross direction lead screw and nut

Directly adjust round nut under the revolving bed until a slight drag is felt while moving the table along the rectangular slides.

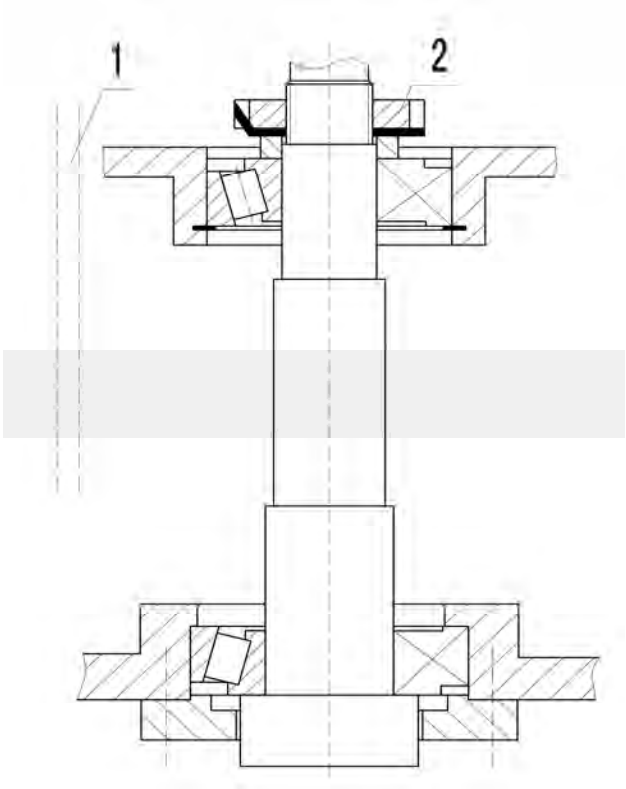


Figure.4

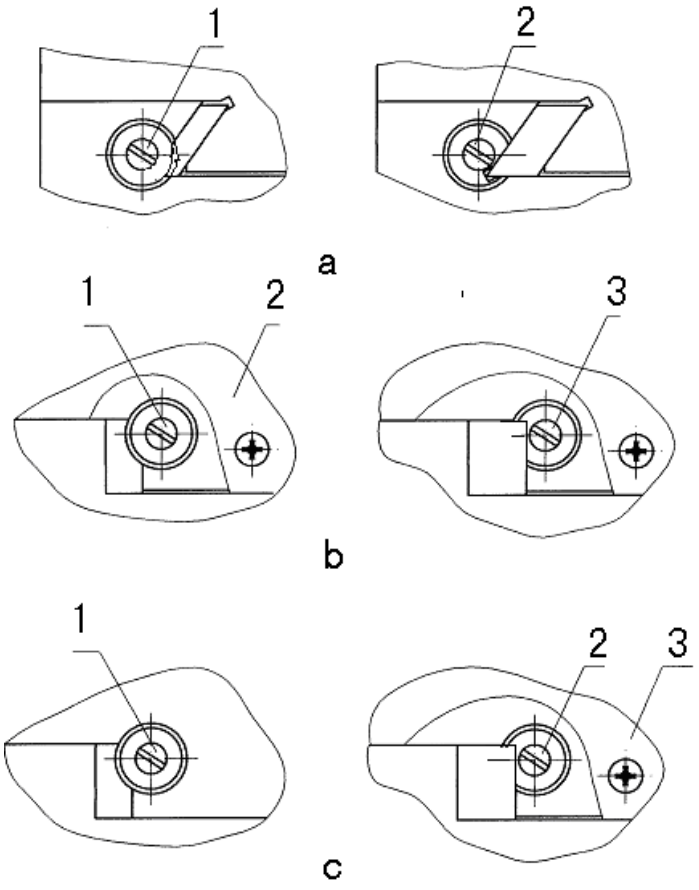
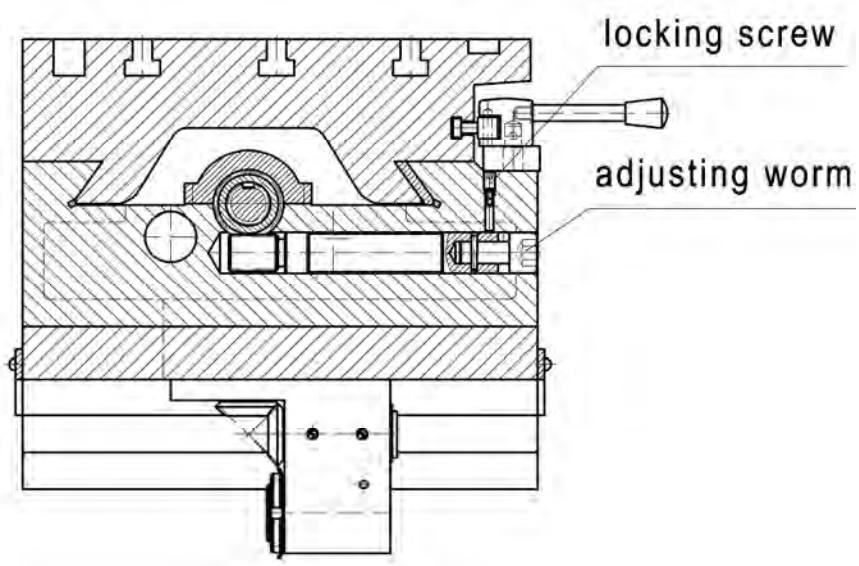


Figure.5



longitudinal direction

Figure.6

8. Cooling and Lubrication of Machine

8.1 Coolant is send to nozzle through pipe. Adjust the nozzle to spray coolant to cutting area and cutter exactly. Coolant is contained in base of the machine.

8.2 Cooling system

Coolant system consists of coolant pump, coolant tank in the base of the machine, orientable and adjustable nozzle and so on.

The valve is used to control the flow of coolant spraying to the cutting area exactly. Coolant is contained in the base of the machine.

MSDS Safety information

Name: JN-8 water-miscible antirust cutting liquid

Ingredients: Non hazardous ingredient

Skin: No

Ingestion: No

Carcinogenicity: No

Appearance and odor: semi- transparence liquid /no odor .

Antirust property : first-class gray cast iron

single piece (drop) 48h eligible; fold pieces 8h eligible

rust test: 55±2℃ cast iron 48h eligible; aluminum 4h eligible

eliminate foam: eligible

PH value: 8.5 extensive test paper

Surface tensility:<40dyn/cm

PB value:>700

μ <0.083

Skin: NO

Ingestion: NO

Carcinogenicity: NO

Effects of exposure: skin: slight irritation.

Explanation of Carcinogenicity: none.

First aid: eyes: flush thoroughly. skin: wash w/soap & water

ingestion: if >1/2 liter (pint), immediately give 1-2 glasses of water & call a physician.

Inhalation: No volatilization at normal temperature, but at high temperature will volatilize hydrocarbon, after inhalation, call a physician.

Extinguishing media: No.

Handling and Disposal

Spill Release Procedures: if a little, absorb on sawdust, or shovel up; If a great lot, prevent spill from entering sewers, drains. soils, then reclaim spill through an approved recycling facility at any government approved waste disposal facility.

1. **Cutting liquid can not be directly drained, it should be reclaimed and processed under the laws and regulations.**
2. **Vapor of coolant which may have a influence to health may arise when the temperature of the tools or the spindle rotation speed is high. So care should be taken that the operating site must kept well ventilated. It is recommended that ventilation equipment should be provided on the operating site.**

8.2 Lubrication

To a great extent, life of machine depends on scientific lubrication.

8.2.1 Lubricating oil must be clean, nonacid, anhydrous and no hard solid and so on.

8.2.2 Main gears adopt automatic force lubricating structure. Reversible cycloidal lubricating pump is installed in the cavity of the column, lubricating pump will start when main motor starts. Lubricating oil is sprayed on gears. Gear box adopts oil immerged splashing lubrication.

In order to lubricate machine well, lubricating oil tank should be cleaned regularly once 3 months at the beginning, once every half a year later.

8.2.3 Please add oil when oil surface is lower than oil scale.

8.2.4 Lubricate lead screws, saddle-table guideway, saddle-knee guideway, knee-column guideway, etc, four times per shift.

8.2.5 Bearing cover of milling holder of hanger adopts dribbing automatic lubrication, should often add oil and clean it.

Other parts which need lubrication have mounted oil cup, every shift must add oil at least 4 times.

MSDS Safety information

Name: N.46 lubricating oil

Health Hazards Data

Ingredients: Non hazardous ingredient

Relative consistency: <1

Inhalation: NO

Skin: NO

Ingestion: NO

Carcinogenicity: NO

Effects of exposure: skin: slight irritation.

Explanation Of Carcinogenicity: none.

First aid: eyes: flush thoroughly. skin: wash w/soap & water

ingestion: if >1/2 liter (pint), immediately give 1-2 glasses of water & call a physician. don't induce vomiting/give anything through mouth. If unconscious, call a physician.

Fire and Explosion Hazard Information

Flash Point Text: 180℃

Extinguishing Media: CO_2, form, dry chemical & water fog.

Hazard: PREVENT RUNOFF FROM FIRE CONTROL OR DILUTION FROM ENTERING STREAMS, SEWERS OR DRINKING WATER SUPPLY.

Physical/Chemical Properties

Solubility in Water: Negligible

Appearance and Odor: Dark amber liquid /no odor.

Reactivity Data

Stability Indicator: YES

Stability Condition to avoid: Extreme heat.
Materials to avoid: Strong oxidizers, combustible material.
Hazardous Polymerization Indicator: NO

Handling and Disposal

Spill Release Procedures: first, cut off fire source, if a little, absorb on sawdust, or Shovel up; If a great lot, prevent spills from entering sewers, drains, soils, then reclaim spill through an approved recycling facility at any government approved waste disposal facility.

Warning: Waste oil should be reclaimed and processed under local laws and regulations.

9. Bearing List of Machine

No.	Name	Model	Specification	Qty	Remark
1	bearing	30314	70×150×38	1	P5
2	bearing	6209	45×85×19	1	
3	bearing	30311	55×120×31.5	1	P5
4	bearing	6309	45×100×25	1	
5	bearing	6308	40×90×23	3	
6	bearing	6211-2RS	55×100×21	1	
7	bearing	6306	30×72×19	1	
8	bearing	6307	35×80×21	1	
9	bearing	6005-2RZ	25×47×12	2	
10	bearing	61807-2RZ	35×47×7	2	
11	bearing	61905-2RZ	25×42×9	4	
12	bearing	6008-2RZ	40×68×15	1	
13	bearing	6005-2RZ	25×47×12	1	
14	bearing	6004-2RZ	20×42×12	1	
15	bearing	32005	25×47×15	2	
16	bearing	61808-2RZ	40×52×7	2	
17	bearing	51110	50×70×14	2	
18	bearing	HKH3024	30×40×24	2	
19	bearing	61910-2RZ	50×72×12	1	
20	bearing	6006-2RZ	30×55×13	1	
21	bearing	51306	30×60×21	1	
22	bearing	7204C	20×47×14	1	

23	bearing	51106	30×47×11	2	
24	bearing	61908	40×62×12	2	
25	bearing	30303	17×47×15.25	1	
26	bearing	6203-Z	17×42×12	1	
27	bearing	32909	45×68×15	1	
28	bearing	6004-Z	20×42×12	3	
29	bearing	51110	50×70×14	2	
30	bearing	HK5020	50×58×20	2	
31	bearing	61904-2RZ	20×37×9	4	
32	bearing	6204-Z	20×47×14	1	
33	bearing	32006	30×55×17	2	
34	bearing	RNA6905	30×42×30	1	
35	bearing	6010-2RS	50×80×16	4	
36	bearing	6008-2RS	40×68×15	1	
37	bearing	6007-2RS	35×62×14	1	
38	bearing	6206-2RS	30×62×16	5	
39	bearing	7009AC	45×75×16	2	
40	bearing	32007	35×62×18	2	
41	bearing	NN3018K	90×140×37	1	P5
42	bearing	30210	50×90×21.75	1	
43	bearing	51116	80×105×19	1	P5

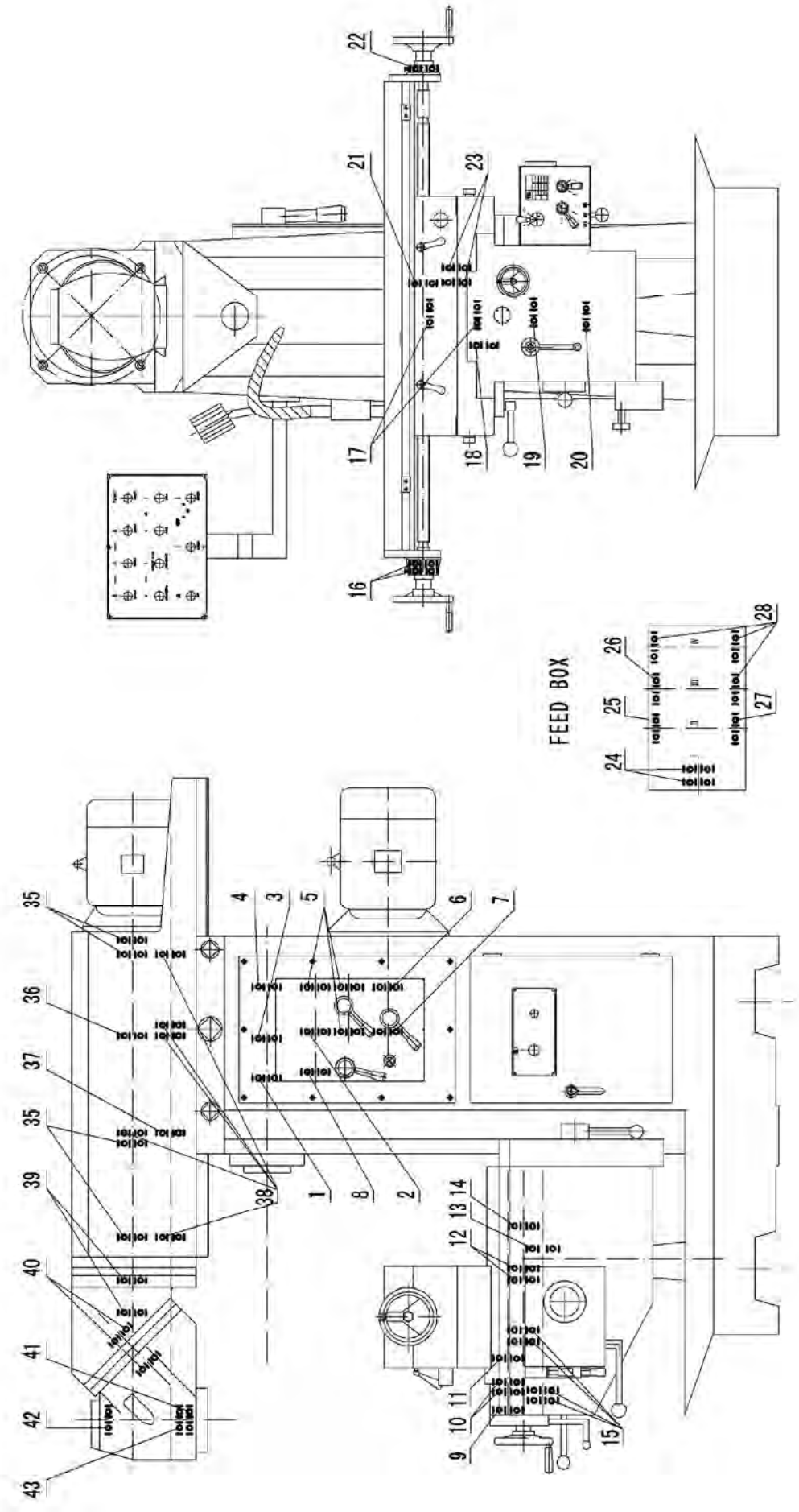


Figure.7

10. Transportation, Installation and Trial Run

10.1 Transportation

The transportation and storage temperature range for the machine tool is between -25℃~ +55℃.

Must load or unload machine according to outer sign on the packing box. Any impact or vibration is prohibited. Must open the box carefully, otherwise it will scratch the paint on the machine surface. After opening the box, check all the accessories according to the packing list, check whether there is something wrong or damaged, please inform dealer or manufacturer in time in order to solve it well, then convey the machine with forklift. Please set a steel wire rope according to Fig.7 when conveying the machine with crane, and place some pads or soft cloth between the machine surface and the steel wire rope, and ask for help if necessary during the transportation .

Caution:

1. The steel-wire rope should not touch machine surface, each hand lever, handle and handwheel. Do put wood block or soft cloth on the interface between the steel-wire rope and the machine edge to avoid damaging the paint.

2. Before conveying the machine with the crane, **move worktable to the front end of the knee, and let the two ends of table on the knee have the same length, tighten the longitudinal & cross locking handles at the same time .**

10.2 Installation

In order to make machine run steadily and keep high machining accuracy, it must be installed on the concrete foundation, which should be built according to the Fig.9.The foundation should be built on solid soil. The machine will be put on the foundation after it is already dry, then fix the bolts on the concrete, must correct it carefully, make sure the tolerance of the machine's levelling is under 0.04/1000 mm in cross and longitudinal directions after tightening the bolts.

10.3 Trial Run

10.3.1 Before trial run, please clean the antirust oil on some parts of the machine carefully, then coat a thin layer of lubricating oil on the outer surface.

10.3.2 Loosen locking handle in three directions (X, Y, Z) of machine before the trial run.

10.3.3 Pour lubricating oil into the gear box and the other lubricating points right, then do an overall check.

10.3.4 Check whether each handwheel and handle of the machine are reliable and flexible.

First start to run idle at the lowest speed for more than 30 minutes, then increase the speed step by step and check whether the handwheels and handles are flexible or reliable.

Caution:Care must be taken when loading or unloading the machine during the transportation.

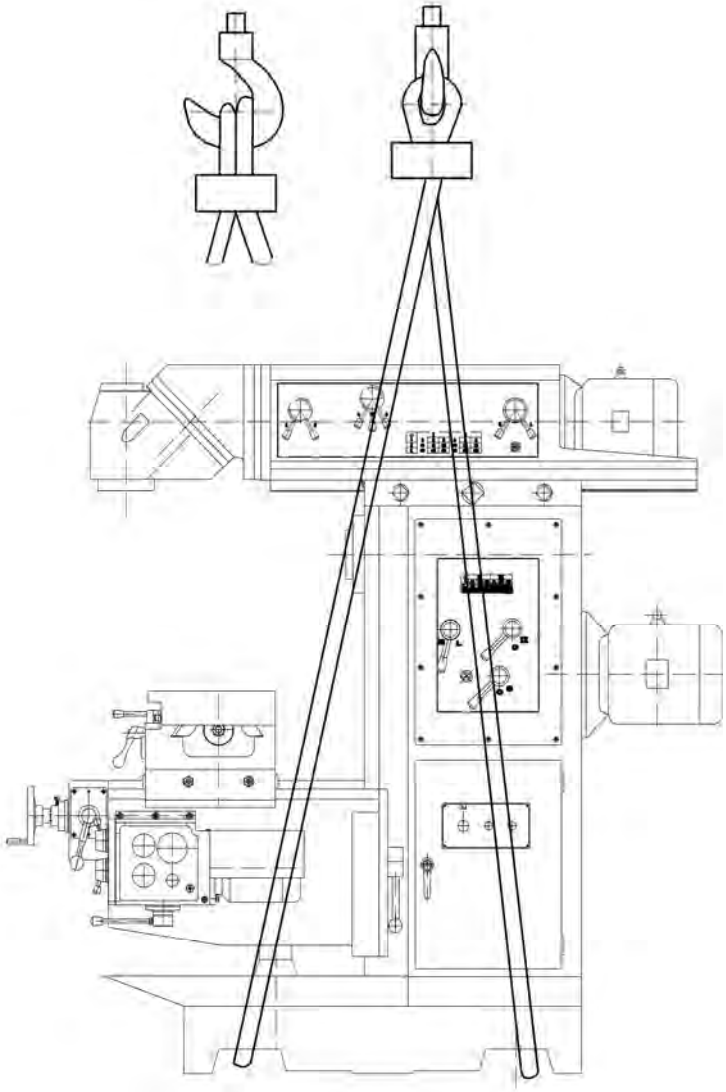


Figure.8

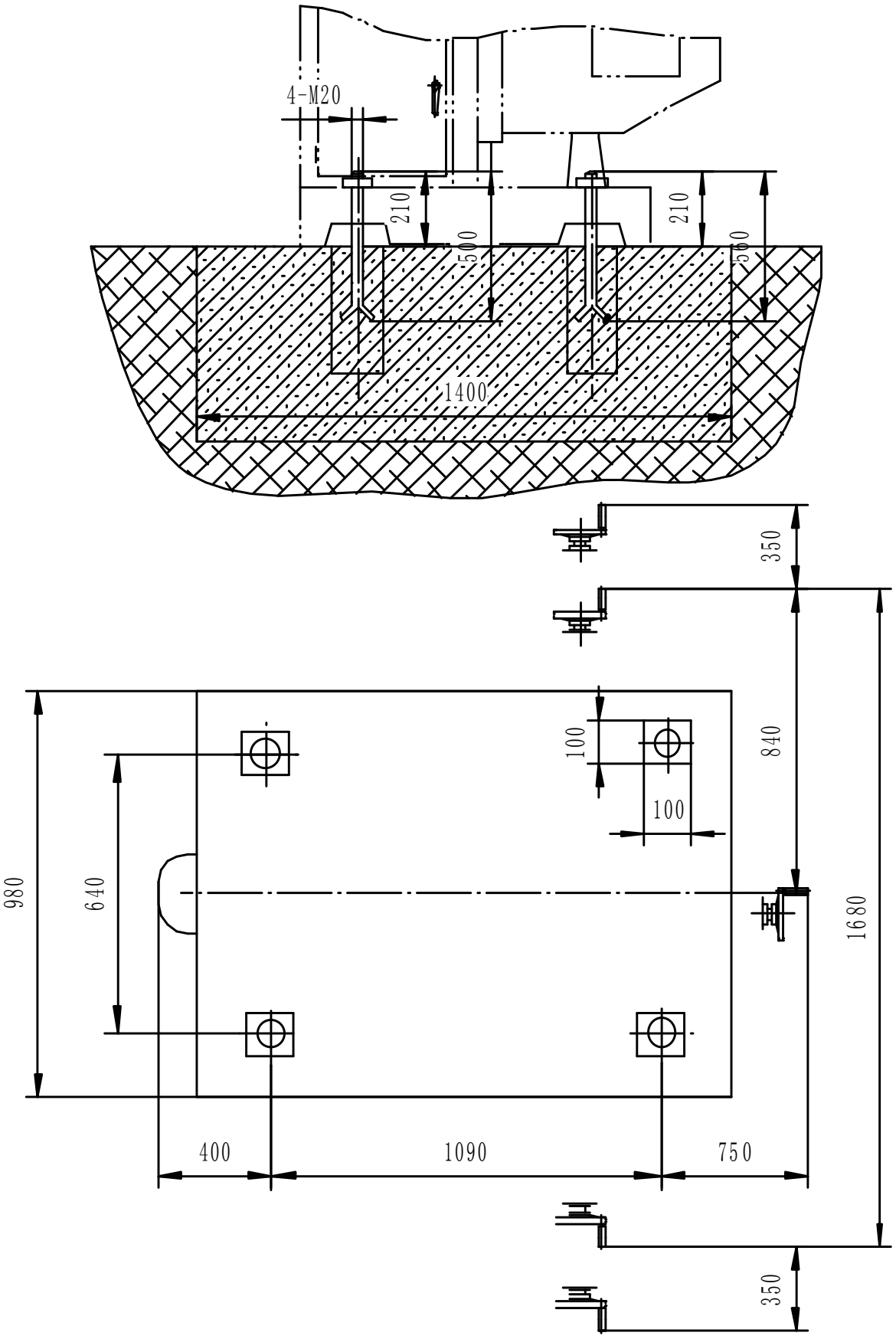

4-M20
210
210
500
560
1400
350
840
100
100
980
640
1680
400
1090
750
350

Figure. 9

11. Maintenance of Machine

11.1 Routine maintenance of machine is very important for accuracy and function of machine.

11.2 Add lubricating oil to oil tank and lubricating site regularly.

11.3 Please often change machining position of workpiece on the table.

11.4 Workpiece and cutter must be tightened before working.

12. Simple Malfunction and Obviation

12.1 Adjust lockinng round nut to eliminate big clearance when the runout of spindle is too big.

12.2 If strange sound happens during the worktable rises or falls, check whether gib is loose, or whether lubricating oil is poor. Please adjust gib and add oil.

12.3 Check whether speed change handles are in place when strange sound happens in gear box, check whether oil is poor, whether shifting fork is damaged. Please adjust the handle’s position, or add oil, or replace the shifting fork.

Caution :

1. **Shut off power supply before repairing the machine.**
2. **The machine should be repaired by professional only.**

13. Electrical System of Machine

Refer to wiring diagram and electrical parts list.

13.1 Power supply: 380V±10%, AC 50±1Hz, 3PH. Switch on main switch after checking power supply.

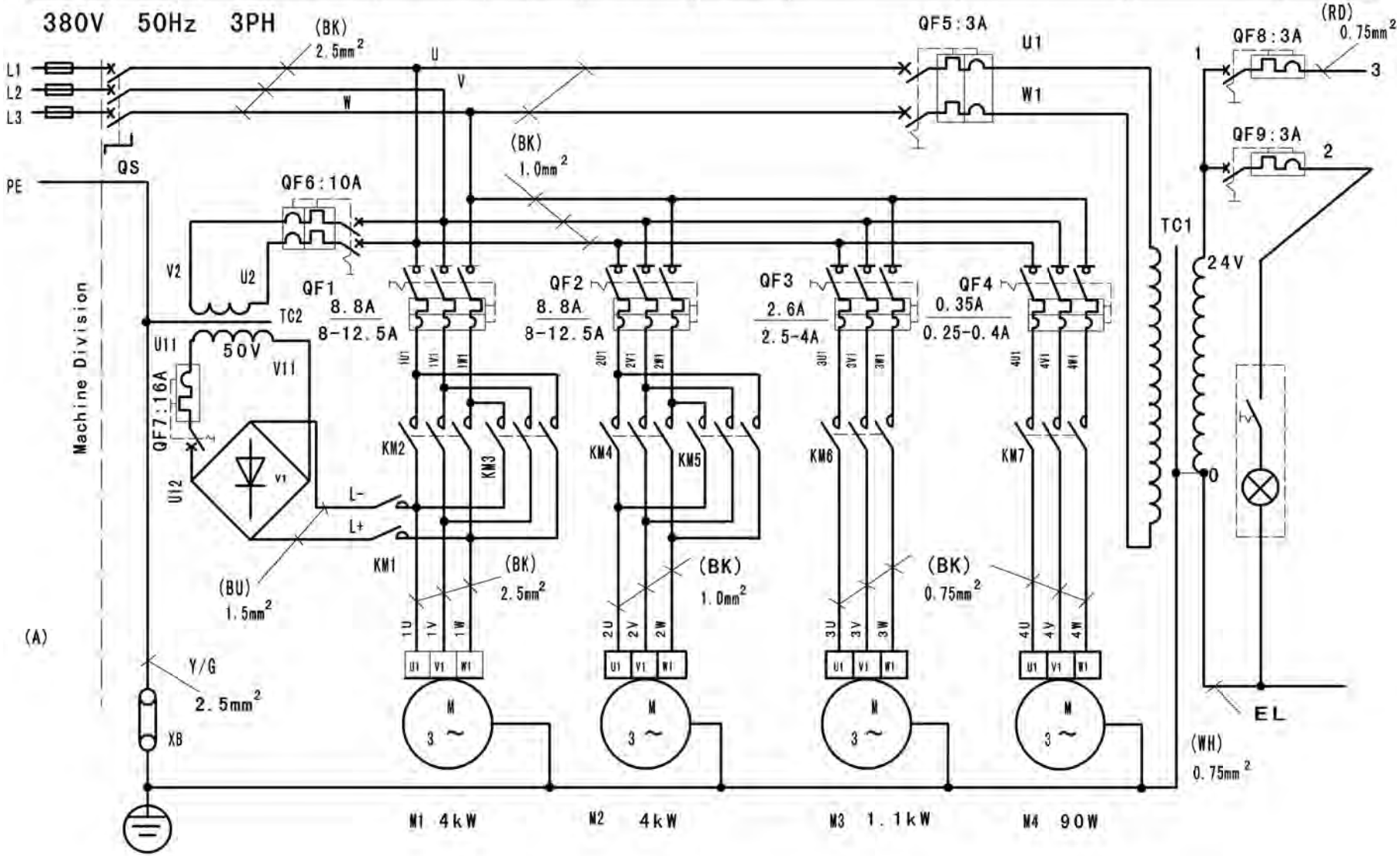
13.2 The machine has many protection functions, such as protection, spindle brake, E-stop and so on.

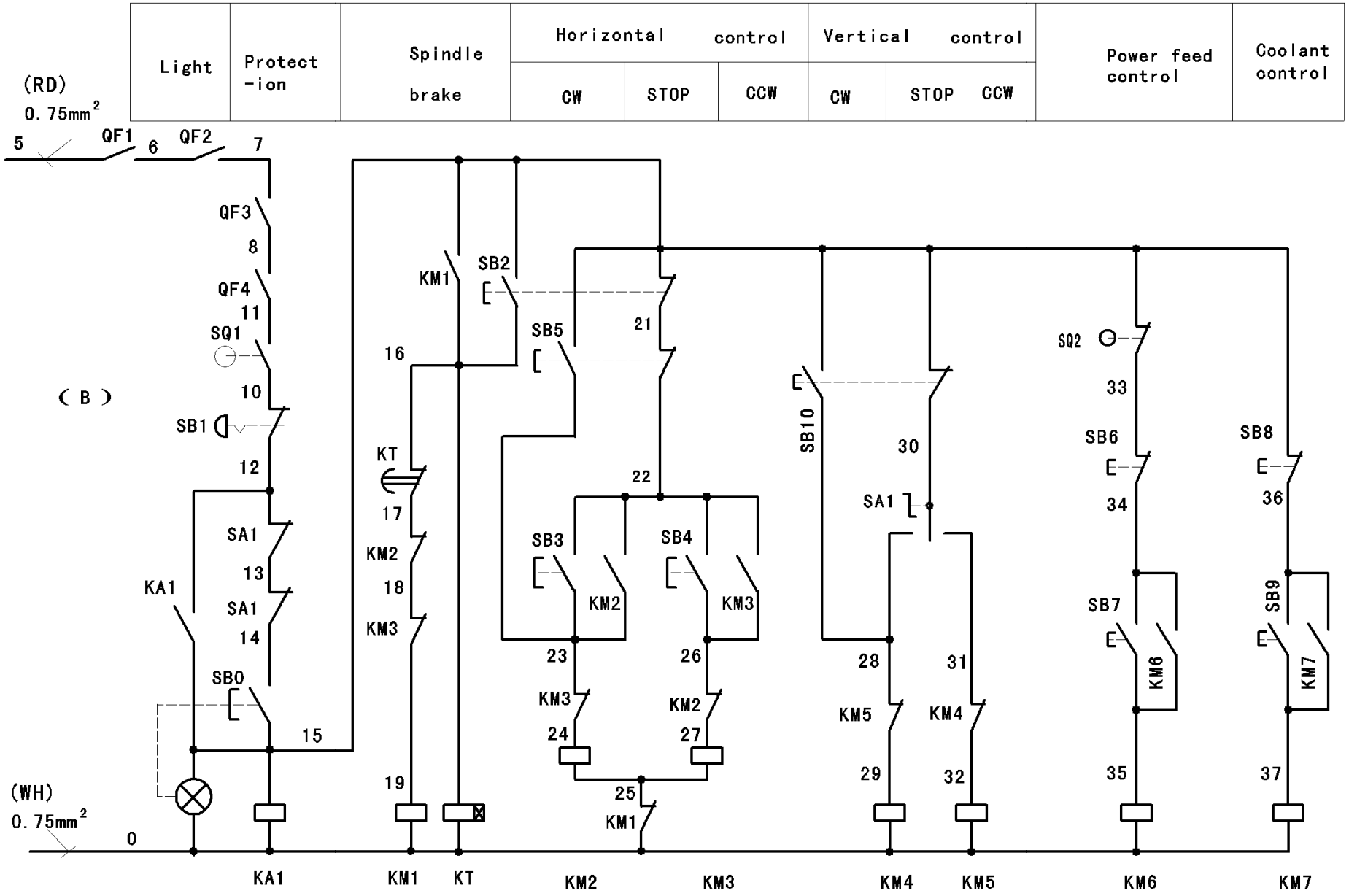
13.3 Press "E-stop" button on the operation panel immediately when malfunction happens, then turn "E-stop" button clockwise after obviating malfunction.

13.4 The machine must be repaired by professional only.

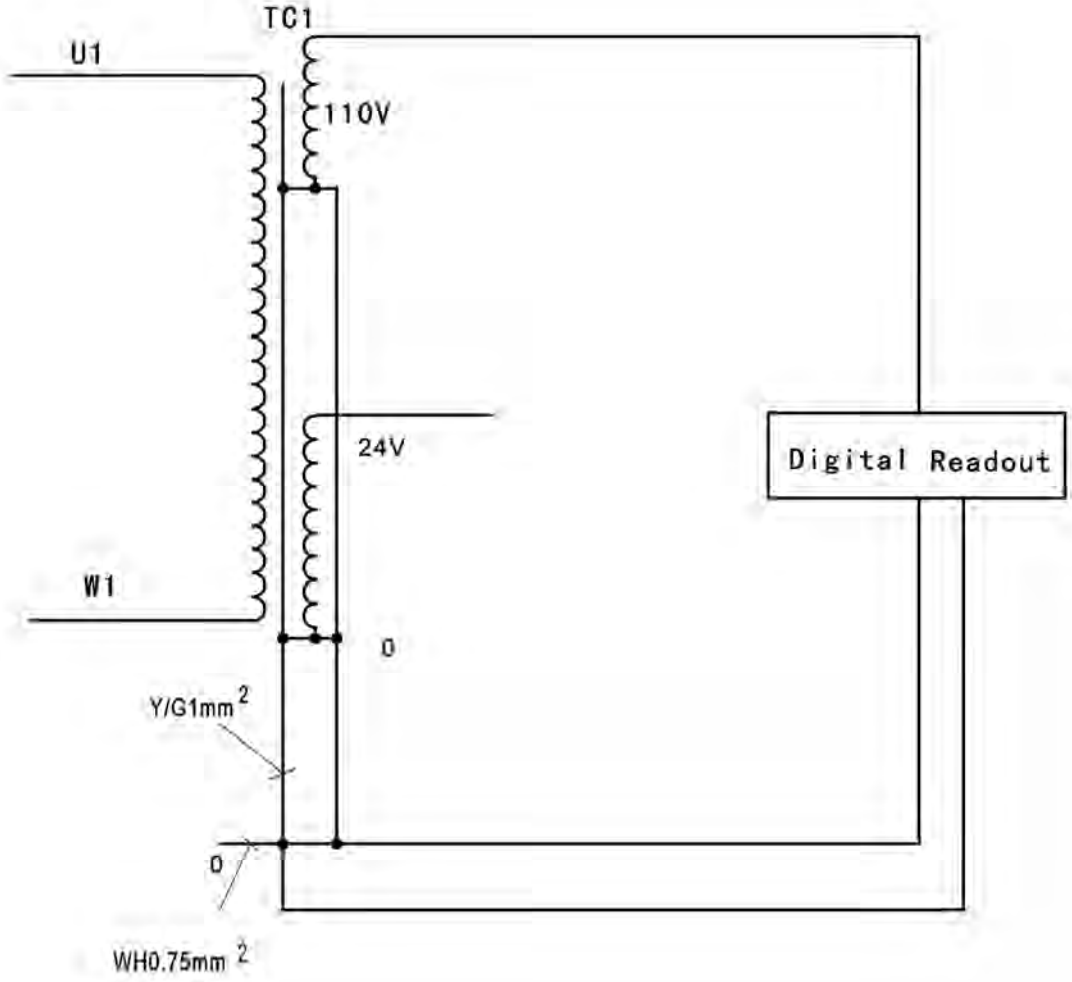
Warning : No repairing on electricity.

Power	Power switch	Protection	Horizontal		Vertical		Power feed	Coolant	Transformer	Light
		Brake	CW	CCW	CW	CCW				




Light
Protect -ion
Spindle brake
Horizontal control
CW
STOP
CCW
Vertical control
CW
STOP
CCW
Power feed control
Coolant control
(RD)
0.75mm²
(WH)
0.75mm²
5
QF1
6
QF2
7
QF3
8
QF4
11
SQ1
10
SB1
12
SA1
13
SA1
14
KA1
SB0
15
0
16
KM1
SB2
SB5
21
KT
17
KM2
18
KM3
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SB3
KM2
SB4
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KM3
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KM2
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SB10
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KM5
29
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KM4
32
SQ2
33
SB6
34
SB7
KM6
35
SB8
36
SB9
KM7
37
KA1
KM1
KT
KM2
KM3
KM4
KM5
KM6
KM7

Transformer		Digital readout



NO.	CODE	NAME	SPECIFICATION	QTY	REMARK
1	M1, M2	motor	Y112M-4 380V 50Hz 3PH 4KW B5	2	
2	M3	motor	Y802-2 380V 50Hz 3PH 1.1kW B5	1	
3	M4	motor	AB-25 380V 50z 3PH 90W	1	
4	QS	switch	JCH-13 32/31	1	
5	QF1, QF2	breaker	JCM5-20(IE:8-12.5A)	2	
6	QF3	breaker	JCM5-20(IE:2.5-4A)	1	
7	QF4	breaker	JCM5-20(IE:0.25-0.4A)	1	
8	QF5	breaker	DZ47-63(2P 3A)	1	
9	QF6	breaker	DZ47-63(2P 10A)	1	
10	QF7	breaker	DZ47-63(1P 16A)	1	
11	QF8, QF9	breaker	DZ47-63(1P 3A)	2	
12	TC1	transformer	JBK5-160 160VA I: 0-380V 0: 0-24V	1	transformer without DRO
13	TC1	transformer	JBK5-200 200VA I:0-380V 0:0-24V(160VA)、0-110V(40VA)	1	transformer with DRO
14	TC2	transformer	JBK5-250 250VA I:0-380V 0:0-45V、50V、55V	1	
15	KA1	relay	LY3 HH63P (AC 24V 50Hz)	1	
16	KM1-KM7	contactor	CJX1-12/22 (AC 24V 50Hz)	7	
17	KT	relay	JS14A-5(AC24V) timing range(0.5-5S)	1	
18	SB0	button	LAY7-10DS/green (AC 24V)	1	
19	SB1	button	LAY7-11MZS/1 red	1	
20	SB2	button	LAY7-11BN/red	1	
21	SB3 SB7 SB9	button	LAY7-10BN/green	3	
22	SB4	button	LAY7-10BN/white	1	
23	SB6 SB8	button	LAY7-01BN/red	2	
24	SB5 SB10	button	LAY3-11/ with inching turn sign "T" green	2	
25	SA1	knob	LAY7-22X/3106	1	
26	V1	rectifier	KBPC2005(Ie:20A)	1	
27	SQ1	micro switch	LXW6-11DL	1	
28	SQ2	travel switch	JLXK1-311	1	
29	EL	working light	JC38B AC:24V 50W	1	
30		lock	MS3001-1-A	1	

14. Adjustment of universal swivel head

14.1 Rotating the head, use positioning round bar by accessory.(Figure.10)

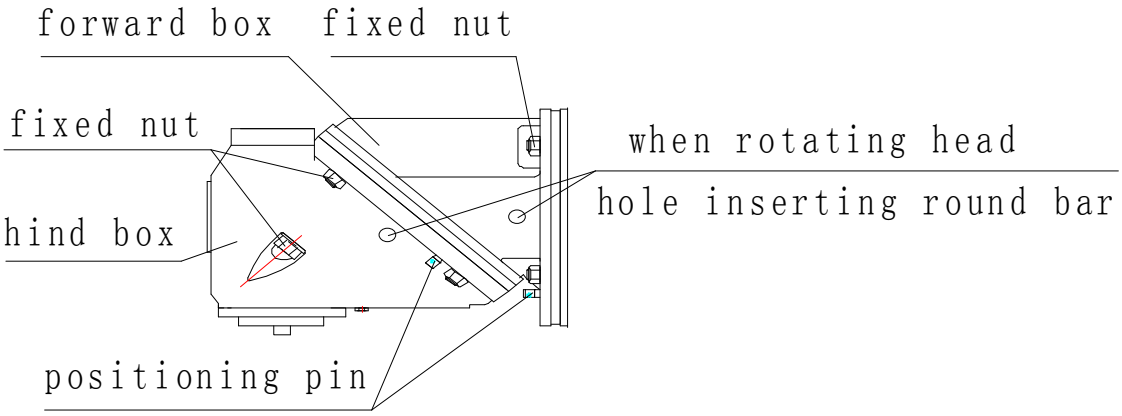


Figure.10

Pull out the positioning pin before rotating the rotation head, then loosen locknut 1-2 pitches，rotate the rotation head in target position, push and insert positioning pin strongly, and clamp the head locknut, so operator can position the head promptly and precisely.

Caution: Don’t unclamp the nut completely, or rotation head will drop from the machine.

It is dangerous to rotate them when unclamping the locknuts of upside head and downside head simultaneously.

14.2 Forward and hind boxes of the swiveling head are all at an angle 0°, the spindle is in the horizontal position. (Figure.11)

Try to install milling tools towards the rootage of milling bar when horizontally milling

14.3 The spindle will be in the vertical position whem forward box is revolved to the angle 180°.

14.4 To expands the machining range, operator will raise the spindle that is in the horizontal position from normal position through revolving hind box an angle 180°. Forward and hind boxes must be positioned precisely with taper pin while the swivel head is in the horizontal or vertical position, which is in order to ensure the vertical and horizontal level precision between the spindle and the table. The taper pin does match special location hole only, don't force the locating pin into the special location hole while it isn't in its position to avoid damaging the junction plate.

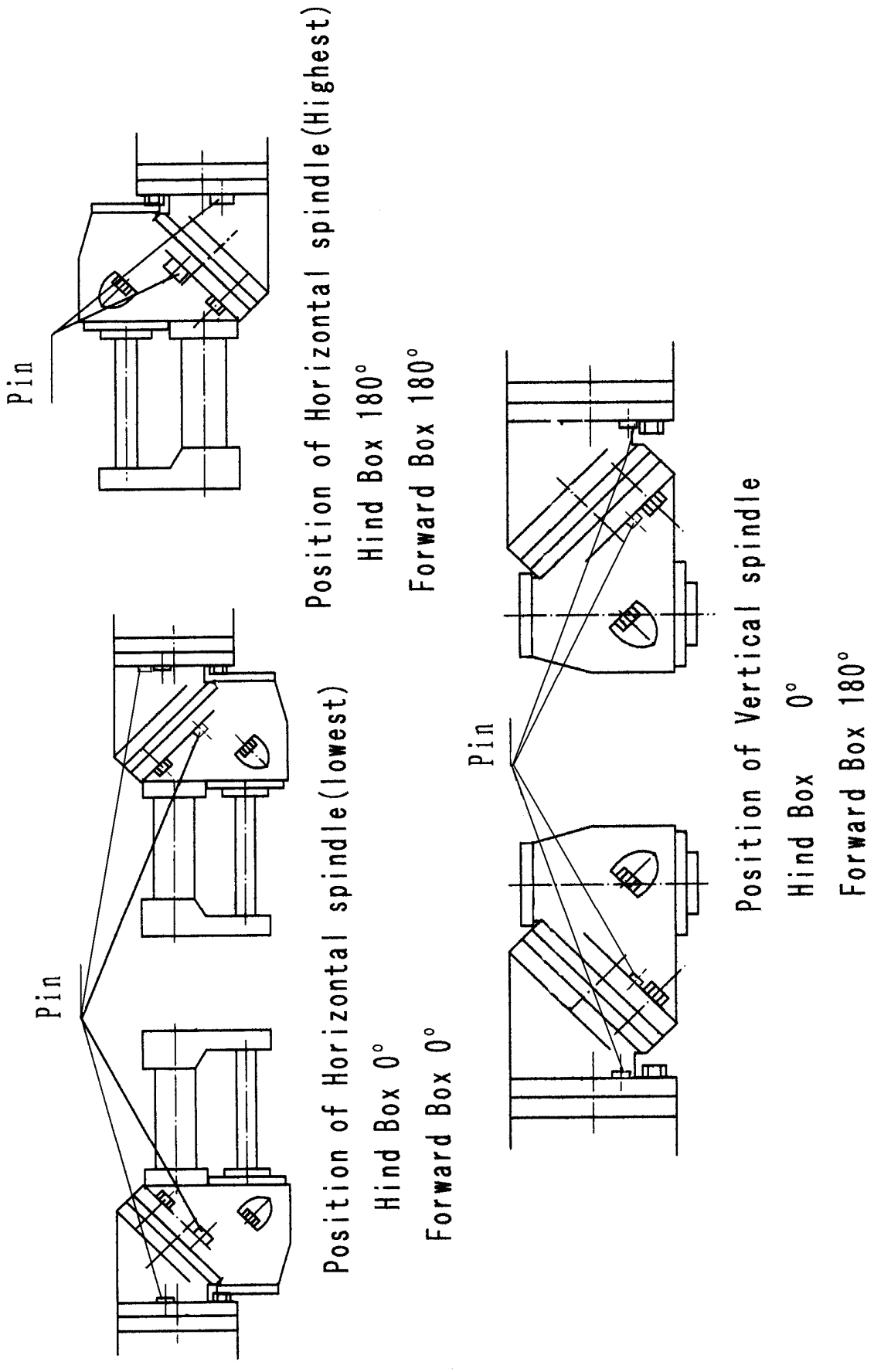

Pin
Position of Horizontal spindle(Highest)
Hind Box 180°
Forward Box 180°
Pin
Position of Horizontal spindle(lowest)
Hind Box 0°
Forward Box 0°
Pin
Position of Vertical spindle
Hind Box 0°
Forward Box 180°

Figure.11

10.2 Right and left tilt of the spindle (Figure.12)

Hind box of the swivel head revolves an angle 90° clockwise or counter-clockwise, which make the spindle work in different position, so the machine will widen the longitudinal processing range.

10.3 Revolve the spindle in horizontal plane. (Figure. 13)

The spindle with the support for milling arbor may revolve in horizontal plane in order to improve the rigidity of the milling arbor. Must revolve forward box and hind box to the different direction when adjusting them. The angle of the spindle is decided by adjustment of forward box and hind Box. The adjustable angle is calculated or checked by the following table.

The formula for calculating is:

$$\cos\beta = 2\cos\theta - 1 \qquad \operatorname{tg}\alpha = \sqrt{2}/2 \operatorname{tg}\frac{\beta}{2}$$

θ ——Angle included between the axis of spindle and cross movement of table.

β — — Forward Box Angle.

α — — Hind Box Angle.

For example:

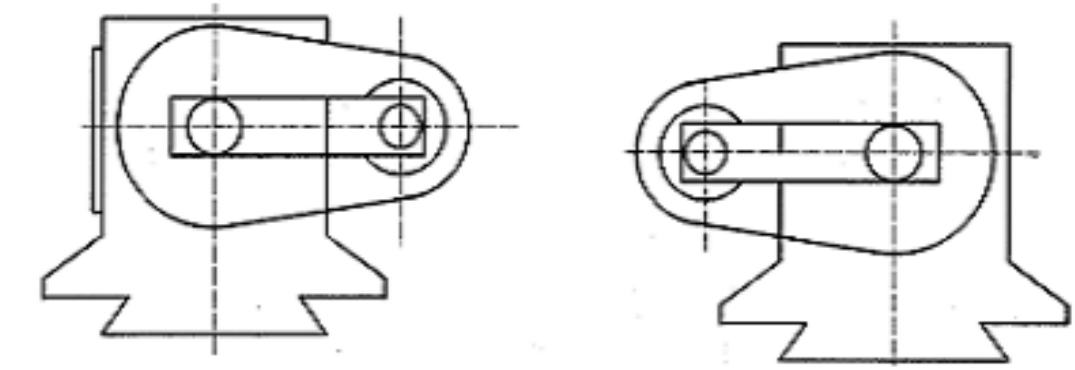
1.cut-45° dextrorotation

Hind Box	α	24° 28' 11" (counterclockwise)
Forward Box	β	65° 31' 49" (clockwise)

2.cut-30° levorotation

Hind Box	α	15° 32' 32" (clockwise)
Forward Box	β	42° 56' 29" (counterclockwise)

.



left tilt of horizontal spindle
hind box 90° (right hand)
forward box 0°

right tlit of horizontal spindle
hind box 90° (levoratation)
forward box 0°

Figure.12

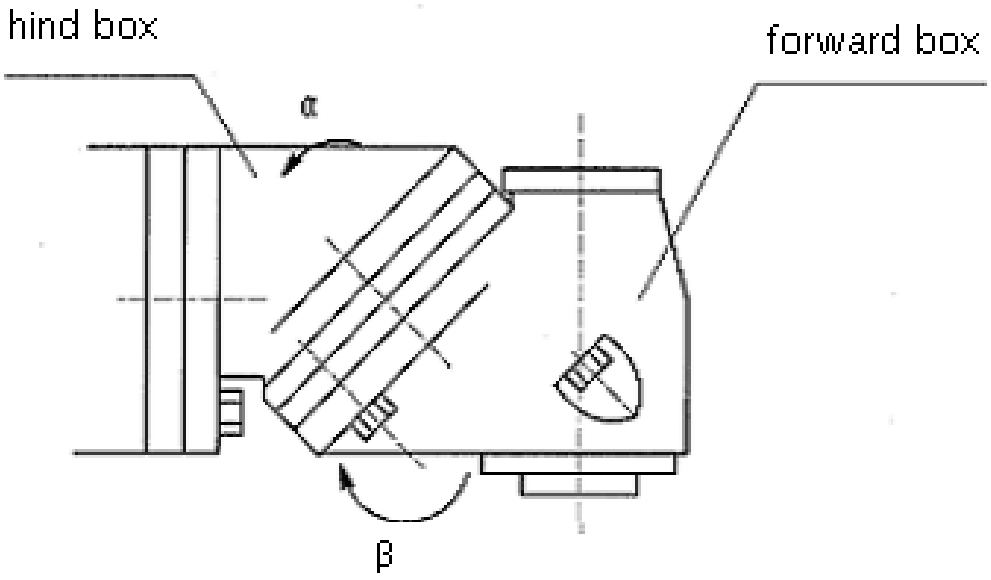


Figure.13

Refer to chart of revolving angles

Spindle angle θ	Forward Box augle β	Hind Box angle α	Spindle angle θ	Forward Box angle β	Hind Box angle α
1°	1°24'51"	0°30'00"	24°	34°11'56"	12°18'20"
2°	2°49'43"	1 00°00'"	25°	35°38'52"	12°48'31"
3°	4°14'35"	1°30'02"	26°	37°05'58"	13°20'53"
4°	5°39'29"	2°00'05"	27°	38°33'17"	13°53'28"
5°	7°04'24"	2°30'09"	28°	40°00'48"	14°26'15"
6°	8°29'21"	3°00'15"	29°	41°28'32"	14°59'17"
7°	9°54'20"	3°30'24"	30°	42°56'29"	15°32'32"
8°	11°19'22"	4°00'35"	31°	44°24'41"	16°06'02"
9°	12°44'28"	4°30'50"	32°	45°53'07"	16°39'48"
10°	14°09'37"	5°01'09"	33°	47°21'50"	17°13'49"
11°	15°35'50"	5°31'32"	34°	48°50'48"	17°48'08"
12°	17°00'08"	6°01'59"	35°	50°20'04"	18°22'44"
13°	18°25'28"	6°32'32"	36°	51°49'38"	18°57'38"
14°	19°50'56"	7°03'10"	37°	53°19'31"	19°32'52"
15°	21°16'29"	7°33'54"	38°	54°49'44"	20°08'27"
16°	22°42'08"	8°04'45"	39°	56°20'17"	20°44'22"
17°	24°07'54"	8°35'42"	40°	57°51'12"	21°20'39"
18°	25°33'46"	9°06'47"	41°	59°22'30"	21°57'20"
19°	26°59'46"	9°38'00"	42°	60°54'10"	22°34'23"
20°	28°25'54"	10°09'21"	43°	62°54'10"	23°11'52"
21°	29°52'11"	10°40'51"	44°	63°58'50"	23°49'48"
22°	31°18'36"	11°12'31"	45°	65°31'49"	24°28'11"
23°	32°45'12"	11°44'20"	46°	67°05'17"	25°07'03"

Spindle angle θ	Forward Box augle β	Hind Box angle α	Spindle angle θ	Forward Box angle β	Hind Box angle α
47°	68°39'15"	25°46'24"	69°	106°27'18"	43°24'55"
48°	70°13'44"	26°26'17"	70°	108°25'08"	44°26'37"
49°	71°48'47"	27°06'42"	71°	110°25'04"	45°30'13"
50°	73°24'24"	27°47'42"	72°	112°27'20"	46°35'50"
51°	75°00'38"	28°28'17"	73°	114°32'08"	47°43'41"
52°	76°37'30"	29°11'30"	74°	116°39'43"	48°53'57"
53°	78°15'02"	29°54'22"	75°	118°30'23"	50°05'52"
54°	79°53'17"	30°37'56"	76°	121°04'29"	51°22'41"
55°	81°32'17"	31°22'13"	77°	123°22'25"	52°41'47"
56°	83°12'04"	32°07'16"	78°	125°44'42"	54°04'30"
57°	84°52'40"	32°53'06"	79°	128°44'53"	55°31'17"
58°	86°34'10"	33°39'47"	80°	130°44'45"	57°02'43"
59°	88°16'35"	34°27'22"	81°	133°24'12"	58°39'30"
60°	90°	35°15'51.8"	82°	136°11'28"	60°22'33"
61°	91°44'28"	36°05'21"	83°	139°08'09"	62°13'04"
62°	93°30'02"	36°55'54"	84°	142°16'26"	64°12'40"
63°	95°17'47"	37°47'33"	85°	145°39'30"	66°23'44"
64°	97°04'48"	38°40'21"	86°	149°22'17"	68°49'50"
65°	98°54'11"	39°34'25"	87°	153°33'02"	71°36'58"
66°	100°45'01"	40°29'49"	88°	158°27'58"	74°56'51"
67°	102°07'23"	41°26'38"	89°	164°49'02"	79°49'34"
68°	104°31'26"	42°24'57"	90°	180°	90°

15. Accuracy Test List

No.	Test item	Tolerance	Measure-ment
1	Levelling of the machine	A: cross 0.04/1000	
		B: longitudinal 0.04/1000	
2	Linearity of the vertical movement of knee	A: cross 0.05/300	
		B: longitudinal 0.05/300	
3	Perpendicularity between vertical guideway and table surface.	A: cross 0.05/300 （a≤90° ）	
		B: longitudinal 0.05/300	
4	Flatness of table	0.06/1000	
5	Parallelism between movement of table and table surface.	A: cross 0.05/300	
		B: longitudinal 0.05/300	
6	Spindle axial kick	0.02	
7	Spindle taper hole axis radial runout	A: near endface of spindle. 0.015	
		B: 300mm to spindle nose 0.04	
8	Parallelism between longitudinal movement of table and datum T-slot..	0.03/300 Max.: 0.06	
9	Parallelism between cross movement of table and spindle axis.	A: on vertical plane 0.05/300 (only down)	
		B: on horizontal plane 0.05/300	
10	Linearity of datum T-slot.	0.03/500 Max.: 0.05	
11	Perpendicularity between spindle axis and table surface.(vertical spindle)	A: cross 0.05/200	
		B: longitudinal 0.05/200	
12	Parallelism between movement of table and surface of table.	A: cross 0.05/300	
		B: longitudinal 0.05/300	
13	Perpendicularity between cross and longitudinal movement of table.	0.04/300	
14	Parallelism between spindle axis and the guideway of ram.	A: on vertical plane 0.05/300 (only down)	
		B: on horizontal plane 0.05/300	
15	Coaxialism between the center hole of hanger and spindle axis.	A:on vertical plane 0.05 (only down)	
		B: on horizontal plane 0.05	
Checker :		Date:	

16. Packing List

No.	Name	Model	Qty
1	Machine		1
2	Milling chuck	ISO50 (8 pieces)	1
3	Inner hexagon spanner	5, 6, 8, 10, 12, 14	each 1
4	Wrench	S21-24, S27-30	each 1
5	Vice	160	1
6	Draw bar	M24（vertical, horizontal）	each 1
7	Middle-sleeve	7:24 ISO50/MS4	1
8	Nut	M16	1
9	Washer	24	1
		16	2
10	Milling arbor	ISO50- ϕ 27 ISO50- ϕ 32	each 1 set
11	Shell end milling arbor	7:24 ISO50- ϕ 32	1
12	Operation manual		1
13	Accuracy test list		1
14	Packing list		1

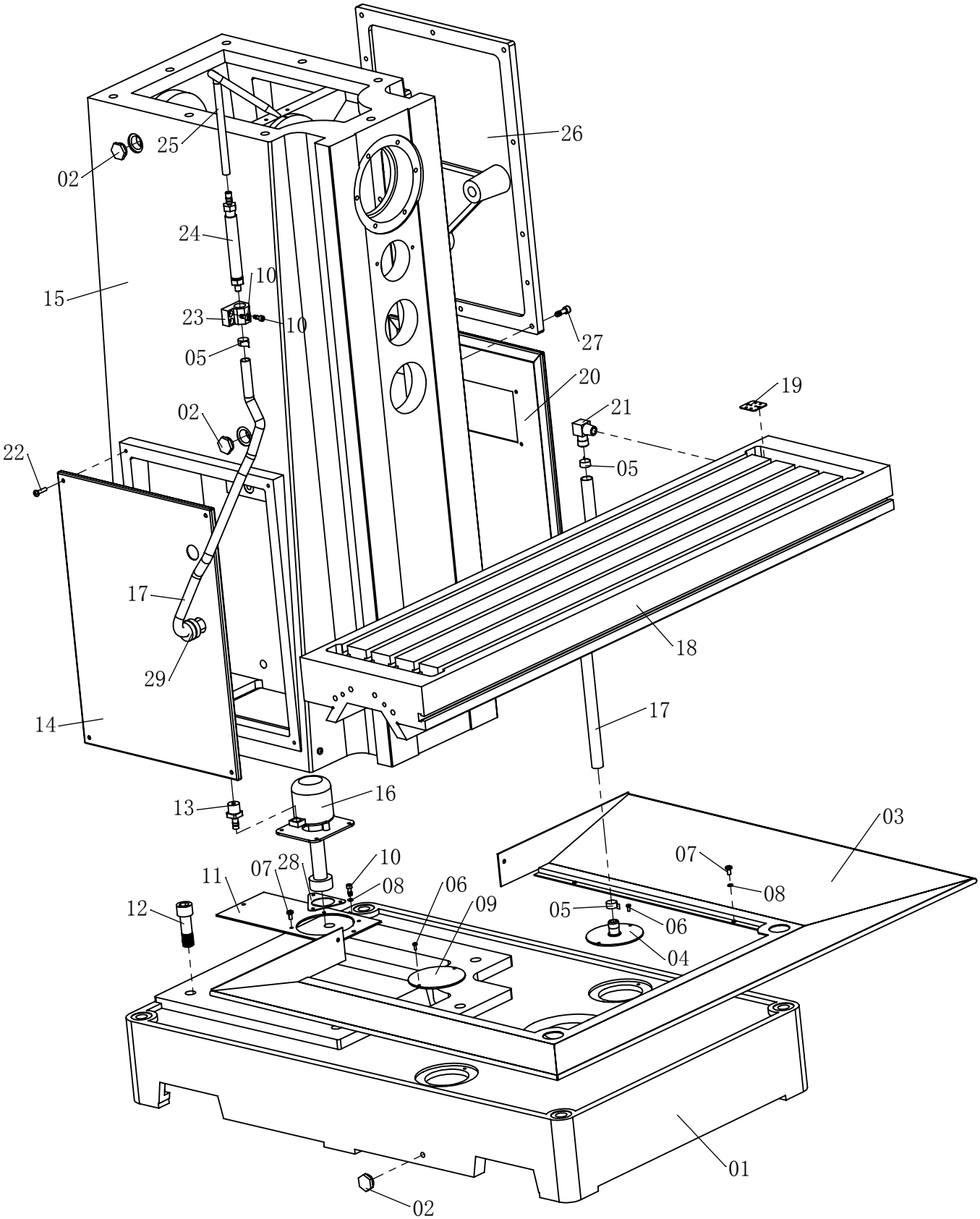
CHESTER
UNIVERAL MILLING MACHINE

VICTORY 360

Parts List and Diagram

Chester UK Ltd
Clwyd Close, Hawarden Industrial Pk
Hawarden, Nr Chester
Flintshire. CH5 3PZ
Tel: 01244 531631
Email: sales@chestermachinetools.com
www.chestermachinetools.com

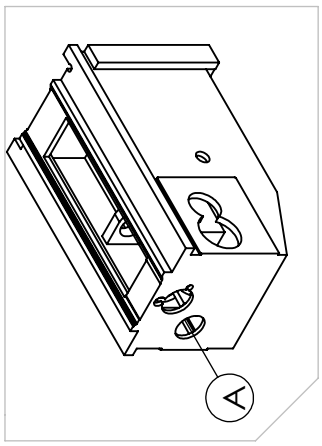
01 Column & base parts

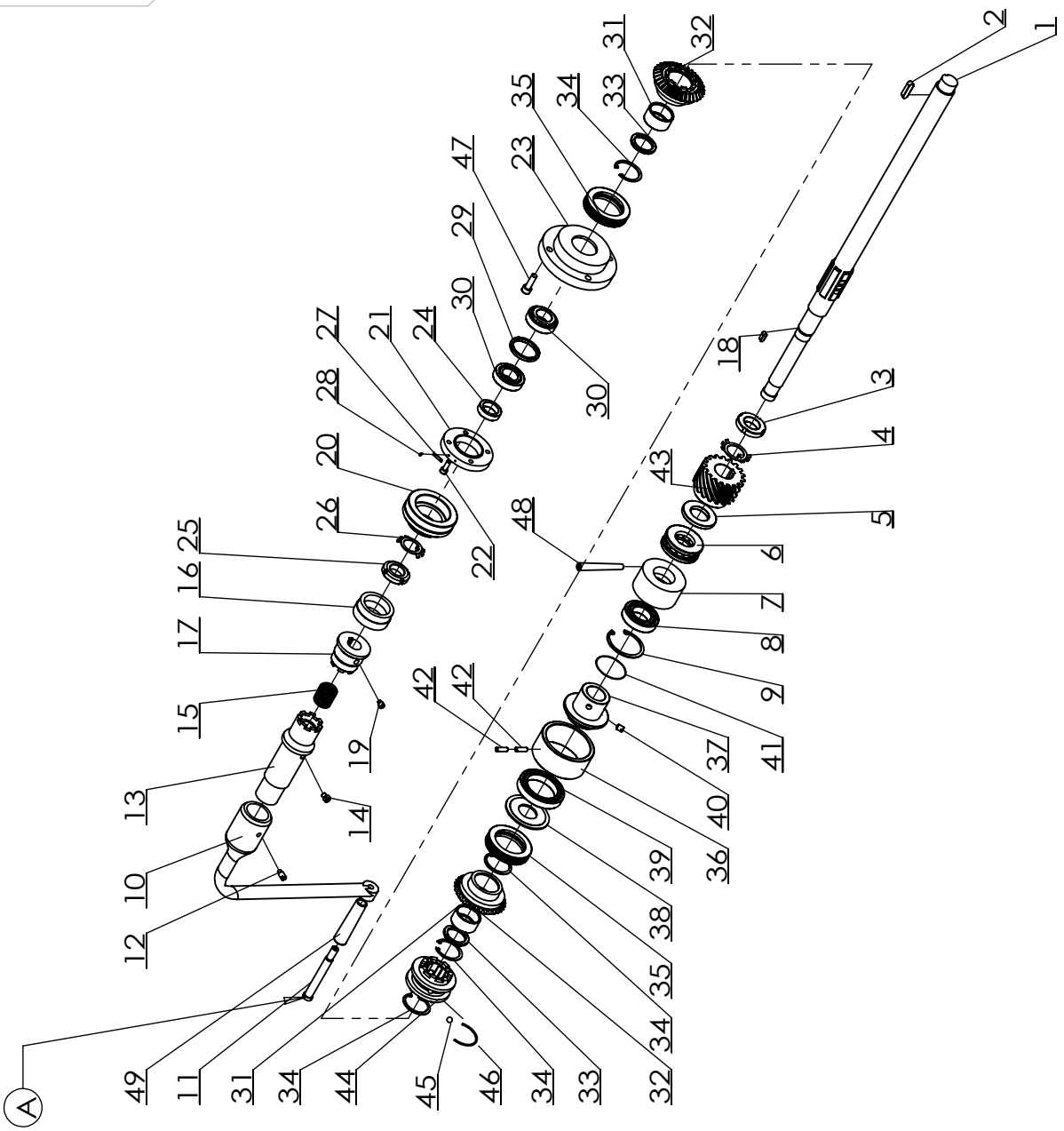


01 Column & base parts

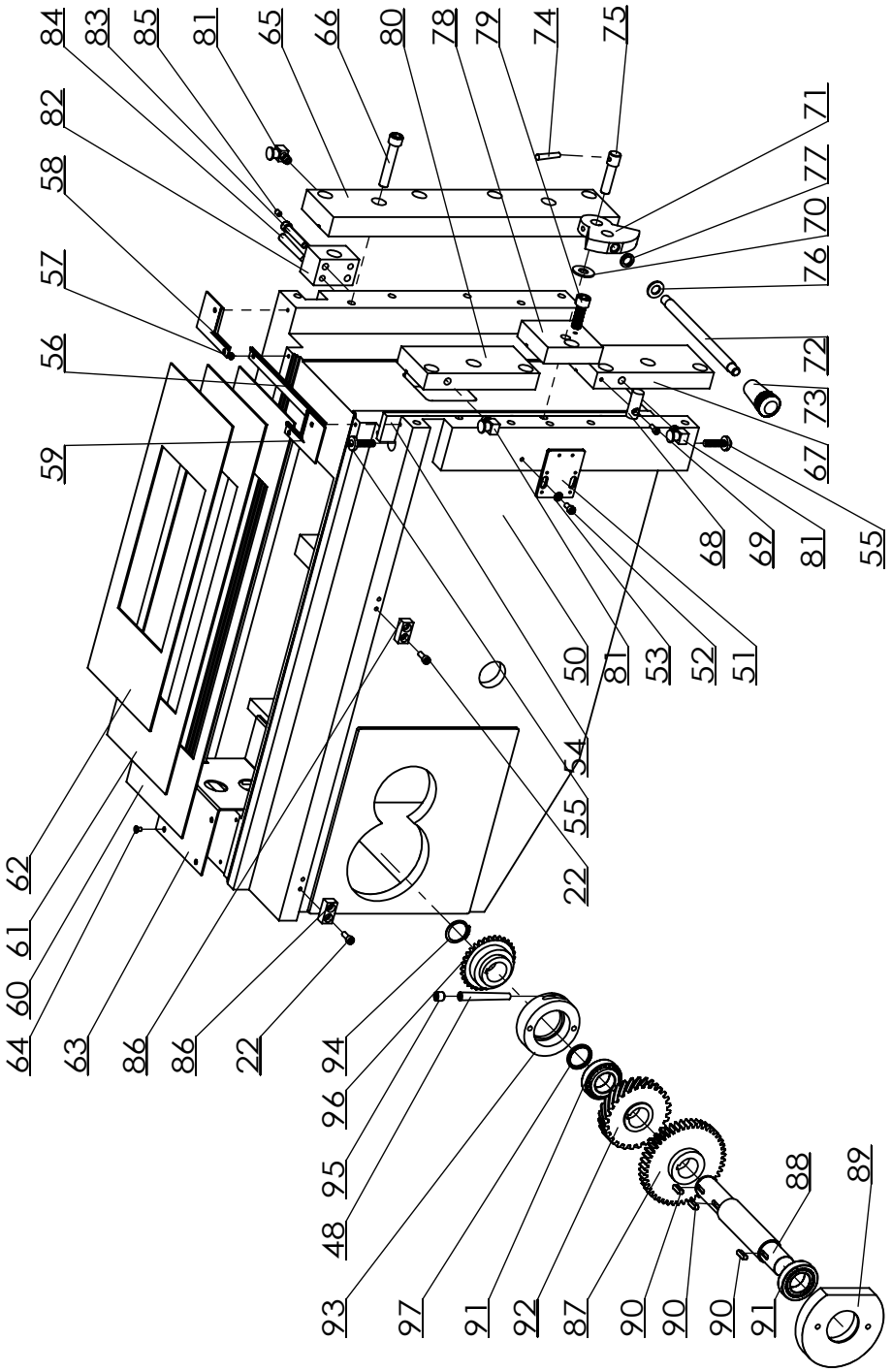
No.	Name	Qty
01001	Base	1
01002	M10×1　Oil plug	3
01003	Tray	1
01004	Pipe joint	1
01005	Φ16-Φ25 Hoop	4
01006	M4×10 Cross recessed countersunk flat head	4
01007	M6×12 Cross recessed countersunk flat head	4
01008	6 Elastic washer	10
01009	Cover board	1
01010	M6×16 Hexagon socket head cap screw	2
01011	Cooling pump block	1
01012	M20×65 Hexagon socket head cap screw	5
01013	Pipe joint	1
01014	Door	1
01015	Column	1
01016	90W　Cooling pump	1
01017	Φ20×1500　PVC pipe	1
01018	Worktable	1
01019	Cover board	1
01020	Door	1
01021	Pipe joint	1
01022	M6×25 Cross recessed pan head screw	1
01023	Fixing base	1
01024	Connecting pipe	1
01025	SR301　Cooling pipe	1
01026	Cover board	1
01027	M8×25 Hexagon socket head cap screw	1
01028	Cover	1
01029	Pipe joint	1

02 Knee Part

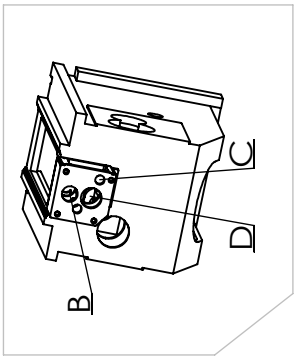

A

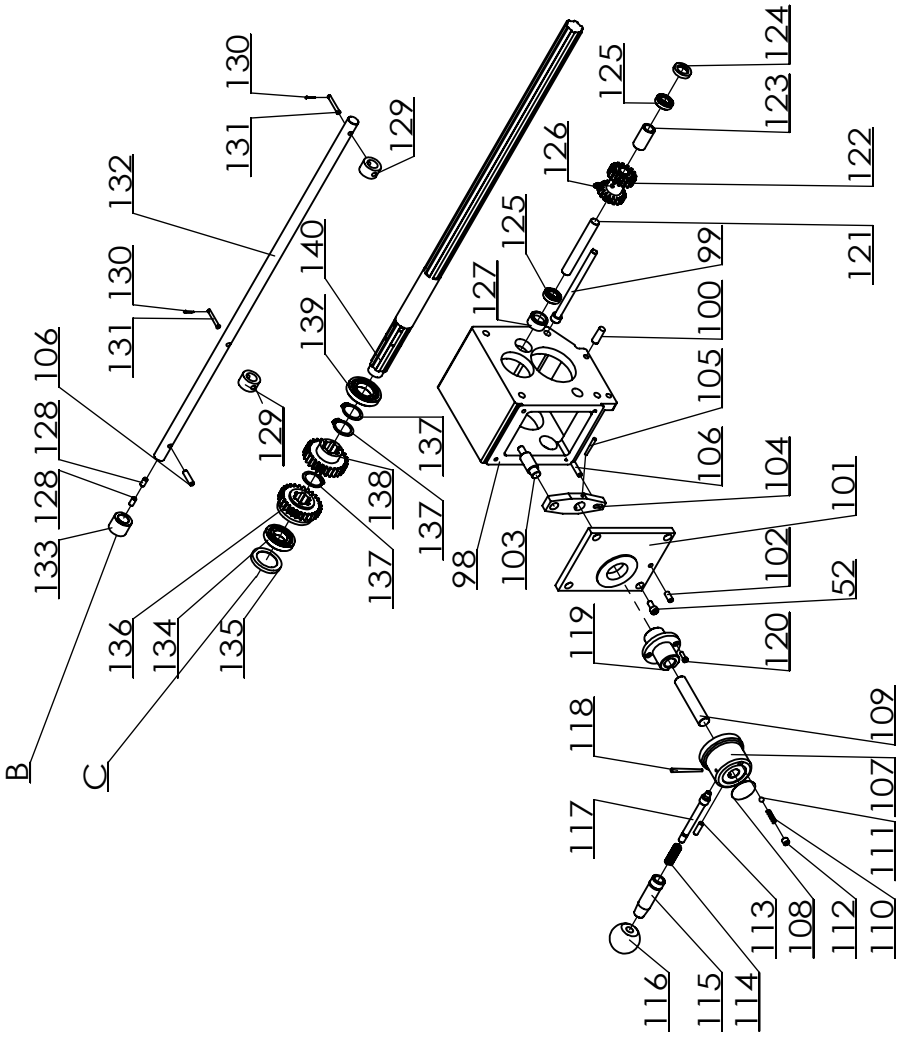

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02 Knee Part

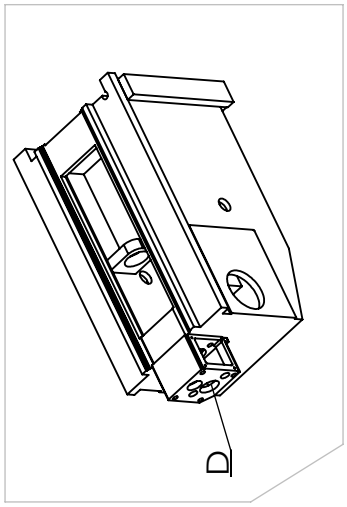

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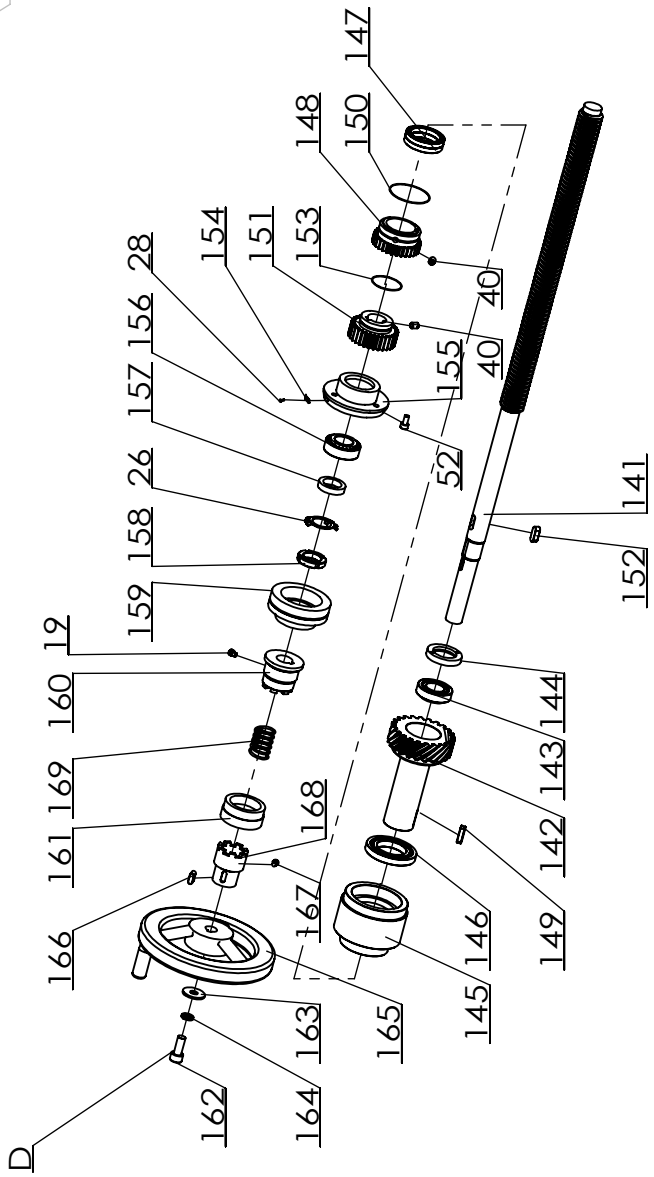
02 Knee Part


C
D
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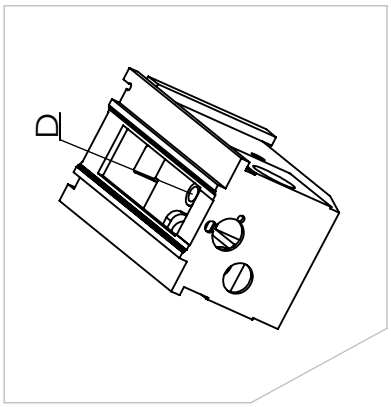

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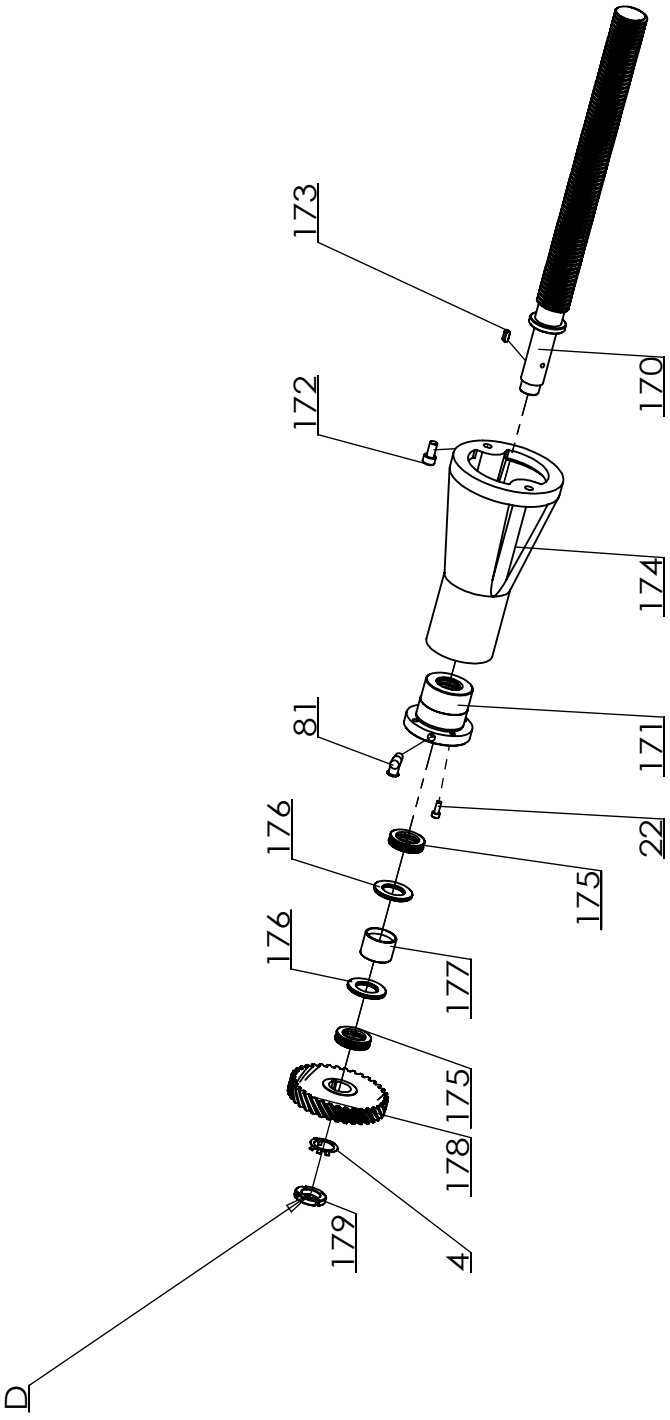
02 Knee Part


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02 Knee Part





02 Knee Part

No.	Name	Qty
02001	Shaft	1
02002	Key	1
02003	Lock nut	1
02004	Lock washer	2
02005	Washer	1
02006	Bearing	1
02007	Sleeve	1
02008	Bearing	1
02009	Retaining ring	1
02010	Crank	1
02011	Grip pin	1
02012	Setscrew	2
02013	Sleeve	1
02014	Setscrew	2
02015	Spring	1
02016	Cover	1
02017	Clutch	1
02018	Key	1
02019	Setscrew	2
02020	Dial	1
02021	Flange	1
02022	Bolt	12
02023	Flange	1
02024	Washer	1
02025	Lock nut	1
02026	Lock washer	2
02027	Point	1
02028	Rivet	4
02029	Washer	1

02 Knee Part

No.	Name	Qty
02030	Bearing	2
02031	Bearing	2
02032	Bevel gear	2
02033	Washer	2
02034	Retaining ring	4
02035	Bearing	2
02036	Collar	1
02037	Collar	1
02038	Washer	1
02039	Bearing	1
02040	Setscrew	3
02041	Wire	1
04042	Setscrew	2
02043	Bevel gear	1
02044	Clutch	1
02045	Steel ball	1
02046	Clip ring	1
02047	Screw	4
02048	Taper pin	1
02049	Grip	1
02050	Knee	1
02051	Plate	1
02052	Screw	9
02053	Washer	2
02054	Gib	1
02055	Adjusting screw	2
02056	Chip cover	1
02057	Cross-recessed head screw	6
02058	Chip cover	1

02 Knee Part

No.	Name	Qty
02059	Chip cover	3
02060	Chip cover	1
02061	Chip cover	1
02062	Chip cover	1
02063	Cover	1
02064	Screw	4
02065	Pressure plate	1
02066	Screw	2
02067	Pressure plate	1
02068	Stopper	1
02069	Screw	1
02070	Adjusting washer	1
02071	Locking boss	1
02072	Operating handle	1
02073	Handle	1
02074	Taper pin	1
02075	Screw	1
02076	Flat washer	1
02077	Elastic washer	1
02078	Pressure plate	1
02079	Screw	1
02080	Pressure plate	1
02081	Oil cup	6
02082	Bracket	1
02083	Screw	2
02084	Taper pin	2
02085	Oil cup	1
02086	Stopper	2
02087	Gear	1

02 Knee Part

No.	Name	Qty
02088	Shaft	1
02089	Cover	1
02090	Key	3
02091	Bearing	2
02092	Gear	1
02093	Collar	1
02094	Shaft retaining ring	1
02095	Setscrew	1
02096	Bevel gear	1
02097	Washer	1
02098	Box	1
02099	Screw	4
02100	Taper pin	2
02101	Cover	1
02102	Taper pin	2
02103	Shaft	1
02104	plate	1
02105	Taper pin	1
02106	Pin	2
02107	Handle boss	1
02108	Nameplate	1
02109	Shaft	1
02110	Spring	1
02111	Steel ball	1
02112	Setscrew	3
02113	Setscrew	1
02114	Spring	1
02115	Grip	1
02116	Grip ball	1

02 Knee Part

No.	Name	Qty
02117	Lever	1
02118	Taper pin	1
02119	Collar	1
02120	Screw	2
02121	Shaft	1
02122	Double-gear	1
02123	Spacer	1
02124	Washer	1
02125	Bearing	2
02126	Setcrew	1
02127	Cover	1
02128	Setcrew	2
02129	Bush	2
02130	Split pin	2
02131	Pin	2
02132	Shaft	1
02133	Collar	1
02134	Bearing	1
02135	Cover	1
02136	Gear	1
02137	Shaft retaining ring	3
02138	Gear	1
02139	Bearing	1
02140	Spline shaft	1
02141	Lead screw	1
02142	Gear	1
02143	Bearing	1
02144	Collar	1
02145	Sleeve	1

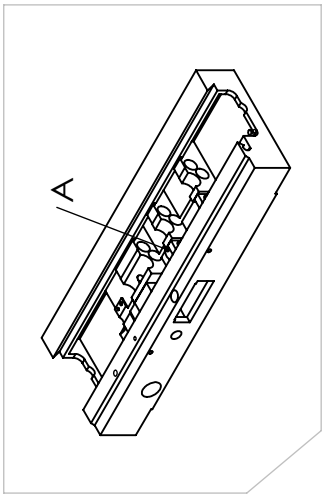
02 Knee Part

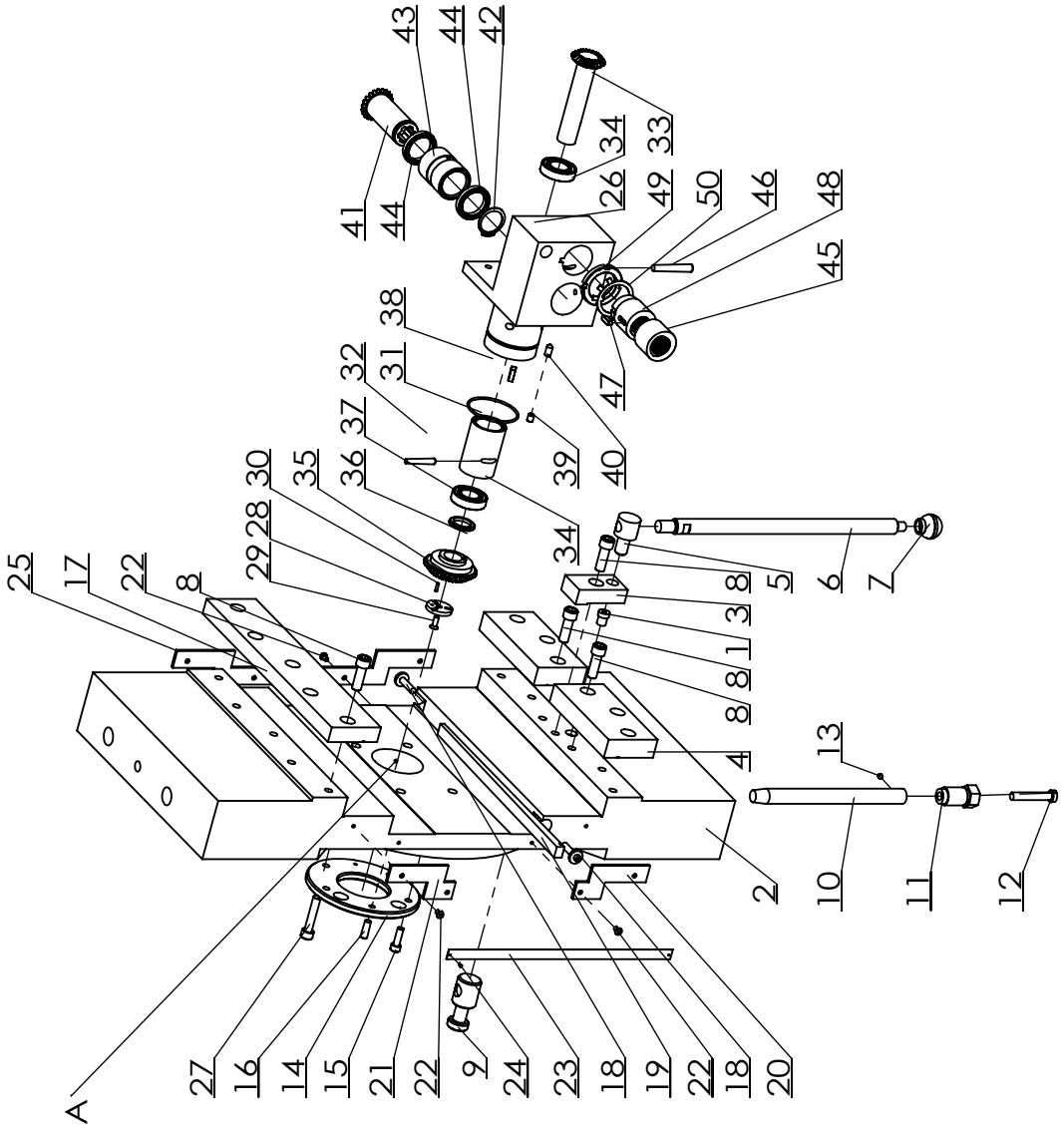
No.	Name	Qty
02146	Bearing	1
02147	Bearing	1
02148	Gaer	1
02149	Key	1
02150	Wire	1
02151	Gaer	1
02152	Key	1
02153	Wire	1
02154	Nameplate	1
02155	Bearing house	1
02156	Bearing	1
02157	Spacer	1
02158	Nut	1
02159	Dial	1
02160	Clutch	1
02161	Cover	1
02162	Screw	1
02163	Washer	1
02164	Elastic washer	1
02165	Handwheel	1
02166	Key	1
02167	Setcrew	1
02168	Clutch	1
02169	Pressure spring	1
02170	Lead screw	1
02171	Nut	1
02172	Screw	2
02173	Key	1
02174	Column	1

02 Knee Part

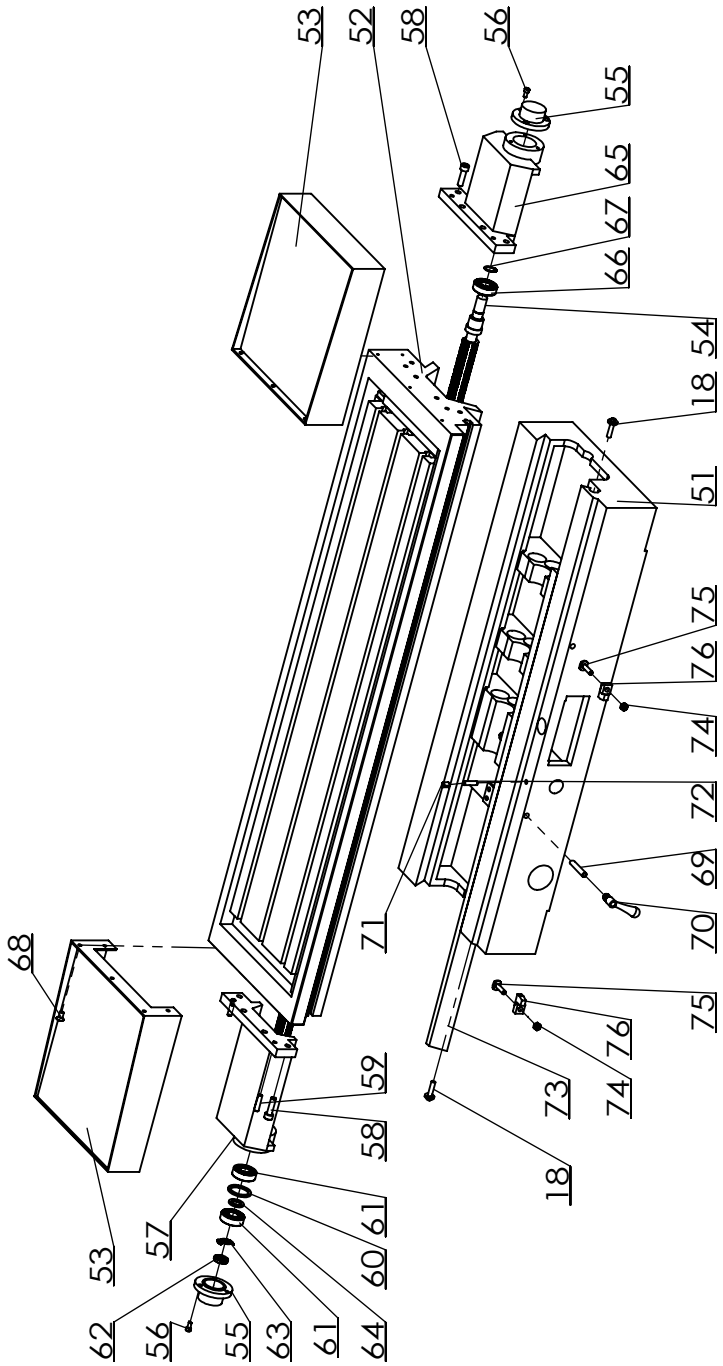
No.	Name	Qty
02175	Bearing	2
02176	Washer	2
02177	Bearing	1
02178	Bevel wheel	1
02179	Nut	1
02180		1
02181		1
02182		1
02183		2
02184		1
02185		1
02186		2
02187		2
02188		2
02189		2
02190		1
02191		1
02192		1
02193		1
02194		1
02195		3
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02197		1
02198		1
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03 Table Part

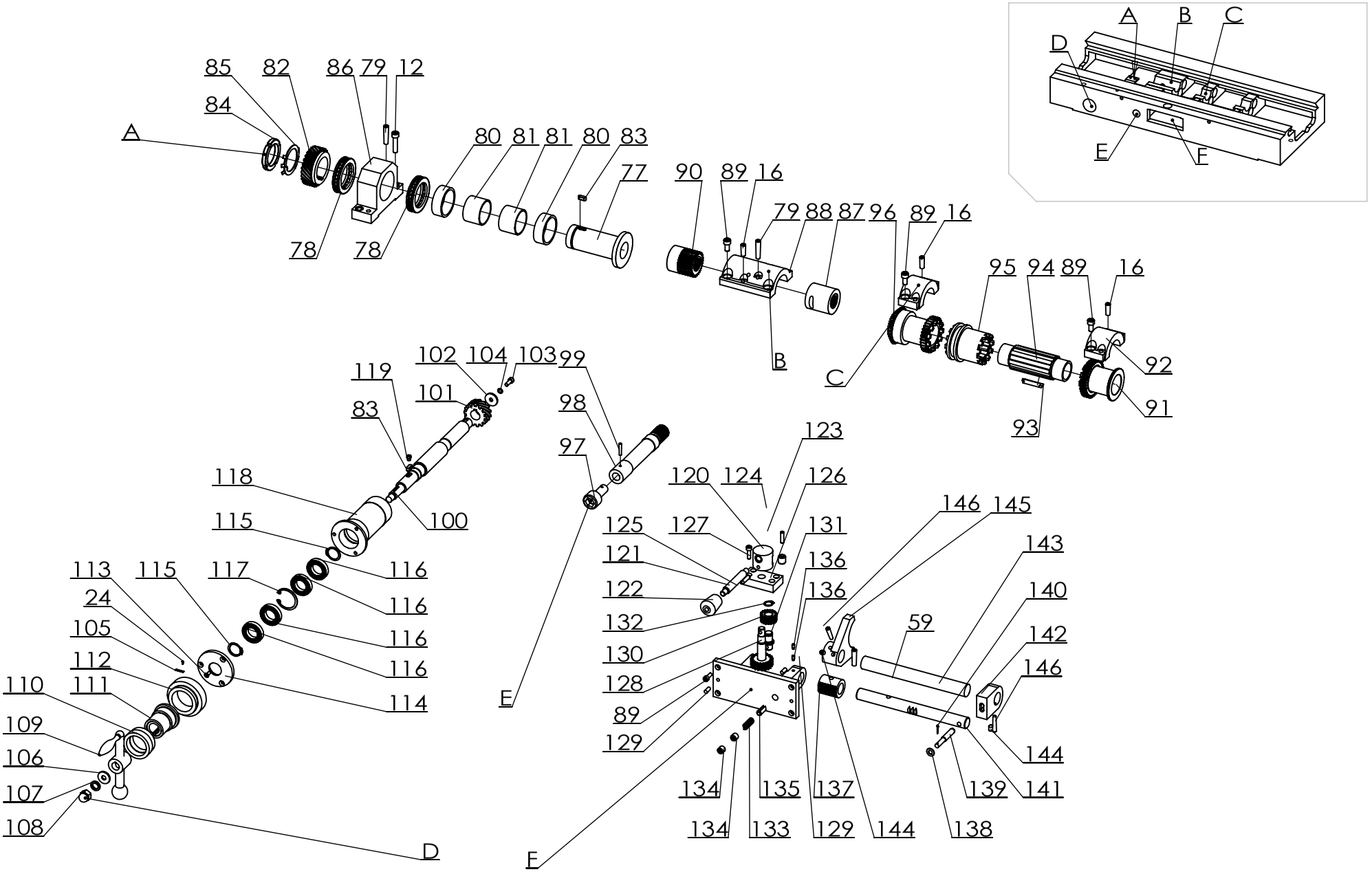




03 Table Part



03 Table Part



103 Table part

No.	Name	Qty
03001	Pin	1
03002	Saddle	1
03003	Pressure plate	1
03004	Pressure plate	2
03005	Clamp bolt	1
03006	Grip	1
03007	Grip ball	1
03008	Screw	11
03009	Claping bolt	4
03010	Bar	4
03011	Screw	4
03012	Screw	4
03013	Setscrew	4
03014	Positioning ring	1
03015	Screw	4
03016	Taper pin	7
03017	Pressure plate	1
03018	Adjusting screw	4
03019	Gib	1
03020	Chip cover	1
03021	Chip cover	1
03022	Screw	8
03023	Gauge	1
03024	Screw	5
03025	Chip cover	1
03026	Supporting bracket	1
03027	Screw	4
03028	Washer	2
03029	Screw	1

03 Table part

No.	Name	Qty
03030	Pin	1
03031	Collar	1
03032	Taper pin	1
03033	Bevel gear	1
03034	Bearing	2
03035	Bevel gear	1
03036	Key	1
03037	Washer	1
03038	O-ring	1
03039	Screw	1
03040	Screw	1
03041	Gearshaft	1
03042	Retaining ring	1
03043	Collar	1
03044	Bearing	2
03045	Nut	1
03046	Taper pin	1
03047	Key	1
03048	Nut	1
03049	Lock nut	1
03050	Lock washer	1
03051	Upper saddle	1
03052	Table	1
03053	Cover	2
03054	Lead screw	1
03055	End cover	2
03056	Screw	6
03057	Left supporting bracket	1
03058	Screw	8

03 Table part

No.	Name	Qty
03059	Taper pin	5
03060	Washer	1
03061	Bearing	2
03062	Lock nut	1
03063	Lock washer	1
03064	Washer	1
03065	Right supporting bracket	1
03066	Bearing	1
03067	Retaining ring	1
03068	Screw	6
03069	Bar	2
03070	Handle	2
03071	Screw	1
03072	Cylinder pin	1
03073	Gib	1
03074	Nut	2
03075	Bolt	2
03076	Stopper	2
03077	Sleeve	1
03078	Bearing	2
03079	Taper pin	4
03080	Bearing	2
03081	Bearing	2
03082	Bevel gear	1
03083	Key	2
03084	Lock nut	1
03085	Lock washer	1
03086	Boss	1
03087	Nut	1

03 Table part

No.	Name	Qty
03088	Cover	1
03089	Screw	12
03090	Nut	1
03091	Gear	1
03092	Gear	1
03093	Key	1
03094	Spline sleeve	1
03095	Sliding sleeve	1
03096	Bevel gear	1
03097	Pin shaft	1
03098	Worm	1
03099	Taper pin	1
06100	Shaft	1
03101	Bevel wheel	1
03102	Washer	1
03103	Screw	4
03104	Washer	1
03105	Point	1
03106	Washer	1
03107	Washer	1
03108	Nut	1
03109	Handle	1
03110	Nut	1
03111	Bush	1
03112	Dial	1
03113	Screw	3
03114	Cover	1
03115	Retaining ring	2
03116	Bearing	4

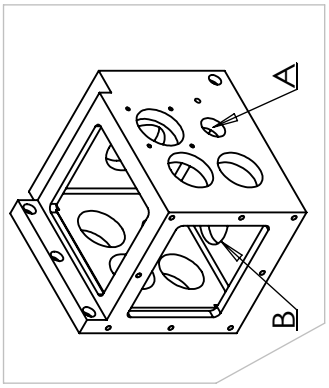
03 Table part

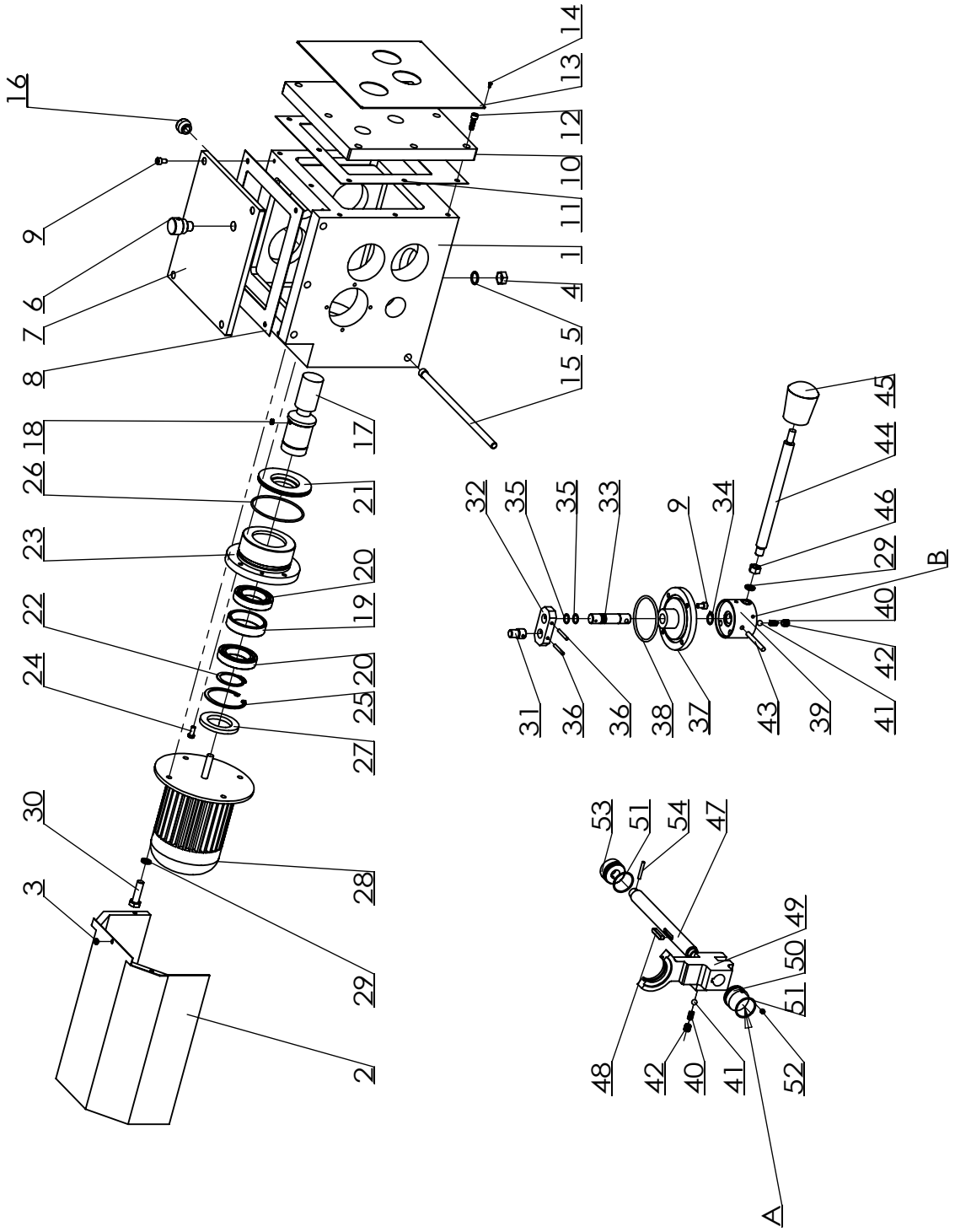
No.	Name	Qty
03117	Retaining ring	1
03118	Bush	8
03119	Screw	1
03120	Handle boss	1
03121	Grip	1
03122	Grip sleeve	1
03123	Wheel	1
03124	Pin	1
03125	Taper pin	1
03126	Pressure plate	1
03127	M6x20 Screw	4
03128	Gear shaft	1
06129	Taper pin	3
03130	Gear	1
03131	Shaft	1
03132	Retaining ring	1
03133	Spring	1
03134	Screw	2
03135	Positioning pin	1
03136	Screw	2
03137	Sleeve	1
03138	Washer	1
03139	Pin	1
03140	Split pin	1
03141	Shaft	1
03142	Block	1
03143	Shaft	1
03144	Screw	2
03145	Shifter	1

03 Table part

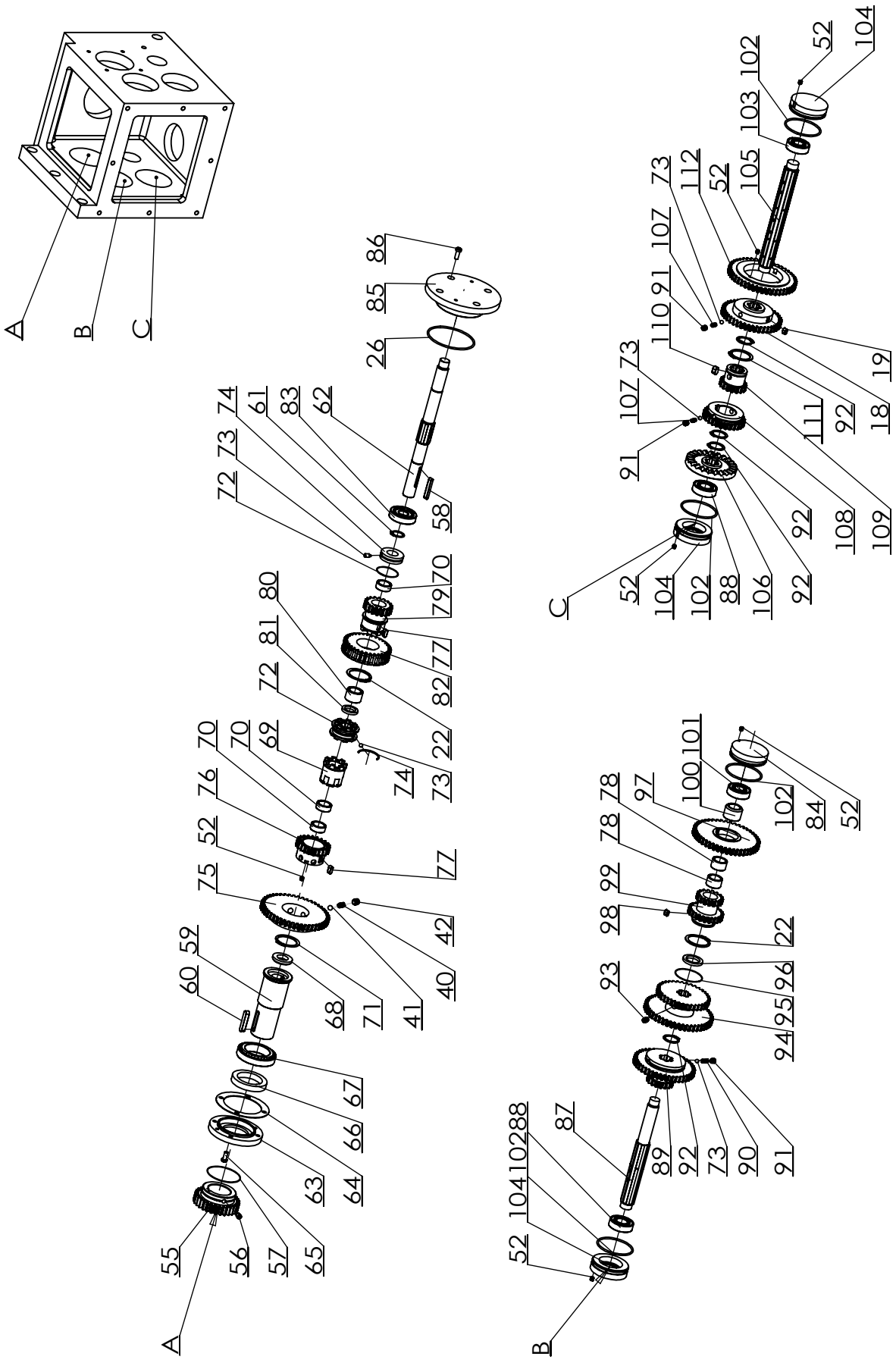
No.	Name	Qty
03146	Taper pin	2
03147		
03148		
03149		
03150		

04 Feed of Table Part

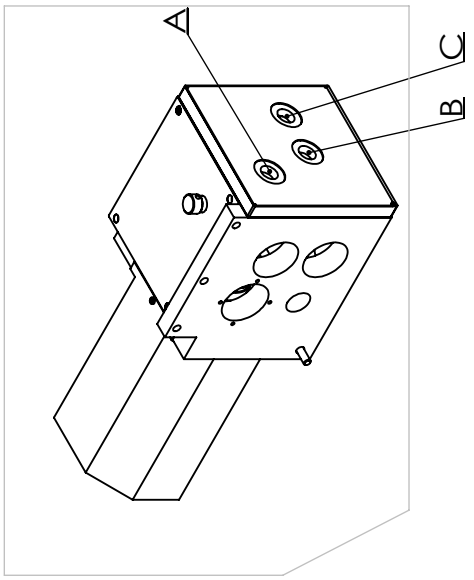


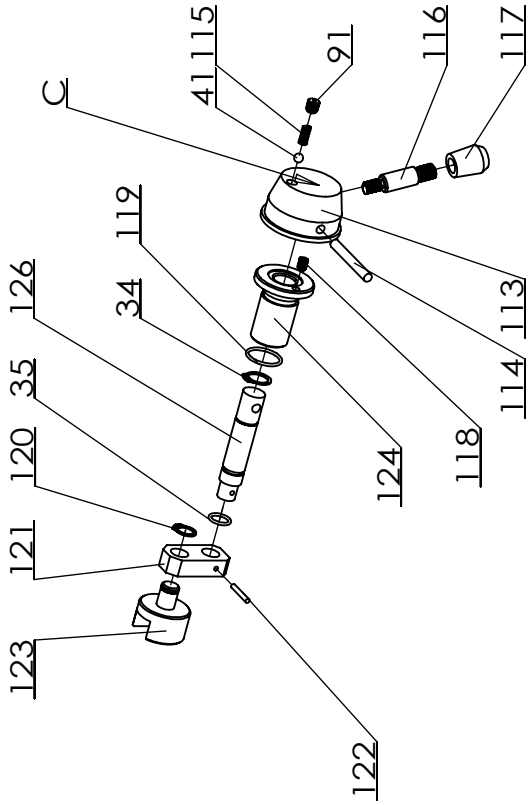


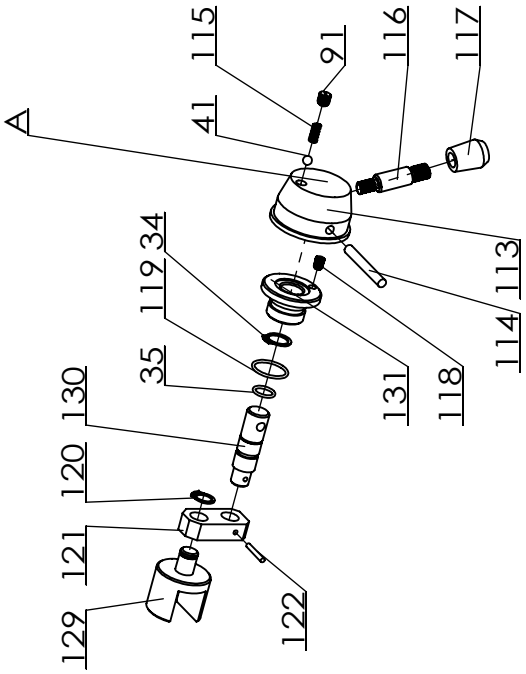
04 Feed of Table Part

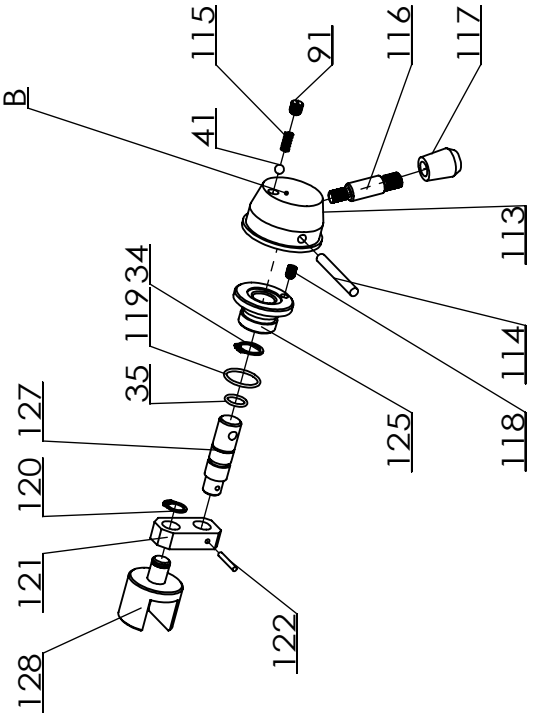


04 Feed of Table Part









04 Feeding box part

N0.	Name	Qty
04001	Feeding box	1
04002	Cover	3
04003	Cross-recessed head screw	1
04004	Oil plug	1
04005	Copper washer	1
04006	Oil plug	1
04007	Cover	1
04008	Washer	1
04009	Screw	8
04010	Cover	1
04011	Washer	1
04012	Screw	8
04013	Nameplate	1
04014	Screw	4
04015	Screw	1
04016	Oil gauge	1
04017	Worm	1
04018	Key	1
04019	Bush	1
04020	Bearing	2
04021	Collar	1
04022	Shaft retaining ring	3
04023	Bearing	1
04024	Screw	4
04025	Retaining ring for hole	1
04026	O-ring	2
04027	Sealing ring	1
04028	Motor	1
04029	Washer	5

04 Feeding box part

N0.	Name	Qty
04030	Bolt	4
04031	Shaft	1
04032	Connecting plate	1
04033	Shaft	1
04034	Shaft retaining ring	4
04035	O-ring	9
04036	Taper pin	2
04037	Collar	1
04038	O-ring	1
04039	Lever boss	1
04040	Spring	8
04041	Steel ball	6
04042	Setscrew	6
04043	Taper pin	1
04044	lever	1
04045	Grip	1
04046	Nut	1
04047	Shaft	1
04048	Key	1
04049	Shifter	1
04050	Plug	1
04051	O-ring	2
04052	Setscrew	17
04053	Plug	1
04054	Taper pin	1
04055	Gear	1
04056	Setscrew	1
04057	Wire	1
04058	Key	1

04 Feeding box part

N0.	Name	Qty
04059	Sleeve	1
04060	Key	1
04061	Shaft retaining ring	1
04062	Shaft	1
04063	Cover	1
04064	Washer	1
04065	Screw	4
04066	Oil seal	1
04067	Bearing	1
04068	Collar	1
04069	Clutch	1
04070	Collar	3
04071	Shaft retaining ring	1
04072	Clutch	1
04073	Steel ball	11
04074	Positioning clip	1
04075	Gear	1
04076	Gear	1
04077	Key	1
04078	Collar	1
04079	Gear	1
04080	Collar	1
04081	Collar	1
04082	Worm wheel	1
04083	Bearing	1
04084	Bearing cover	1
04085	cover	1
04086	Screw	4
04087	Spline shaft	1

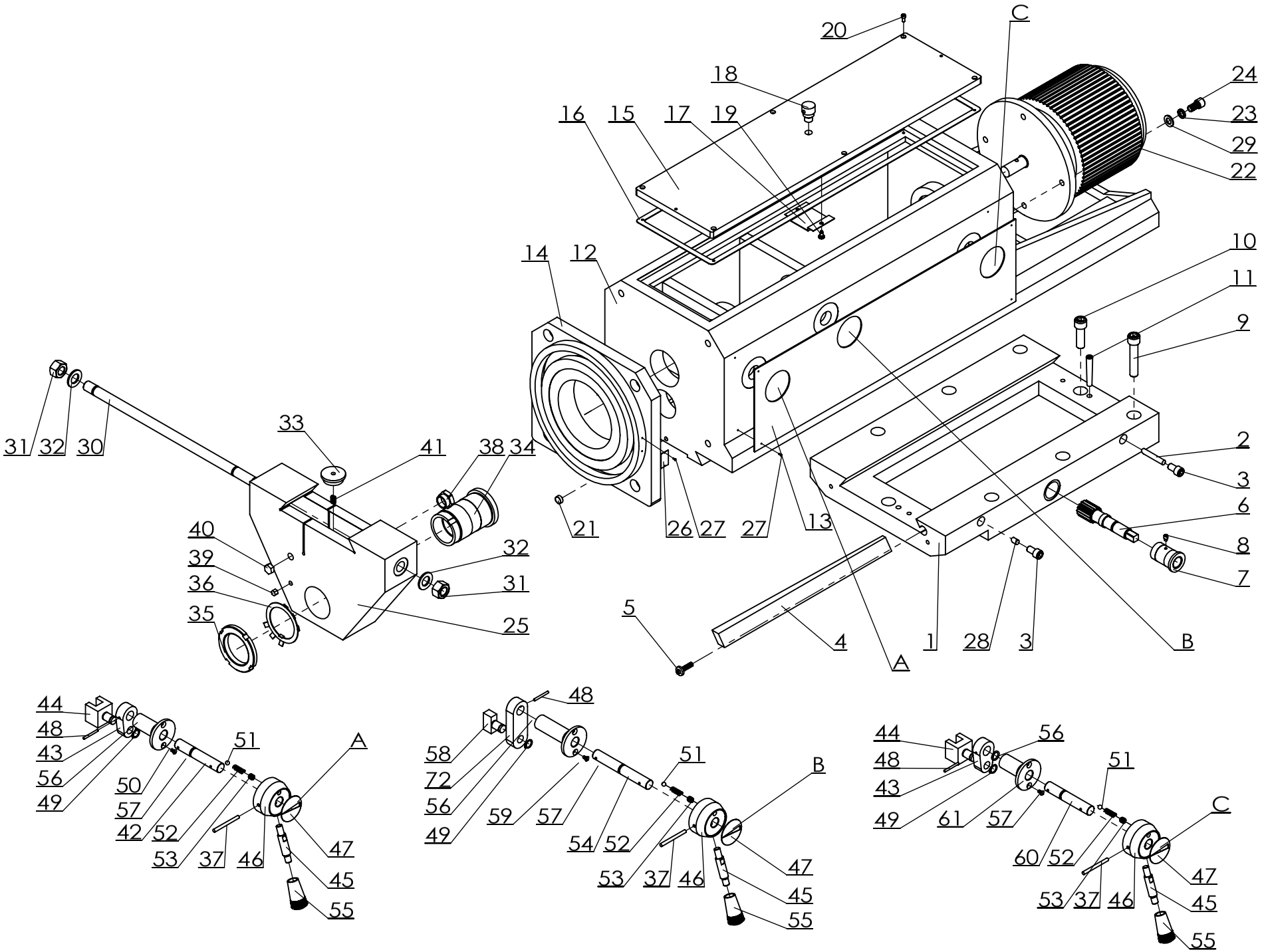
04 Feeding box part

N0.	Name	Qty
04088	Bearing	3
04089	Gear	1
04090	Spring	2
04091	Setcrew	9
04092	Shaft retaining ring	4
04093	Setcrew	1
04094	Gear	1
04095	Wire	1
04096	Washer	1
04097	Gear	1
04098	Key	2
04099	Gear	1
04100	Collar	1
04101	Bearing	1
04102	O-ring	4
04103	Bearing	1
04104	Bearing boss	3
04105	Spline shaft	1
04106	Gear	1
04107	Spring	1
04108	Gear	1
04109	Gear	1
04110	Key	1
04111	Shaft retaining ring	1
04112	Gear	1
04113	Lever boss	3
04114	Taper pin	3
04115	Spring	3
04116	lever	3

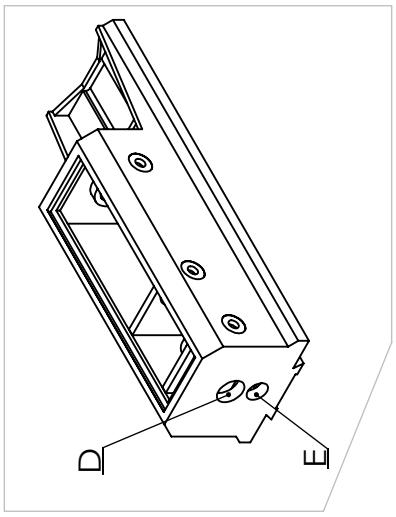
04 Feeding box part

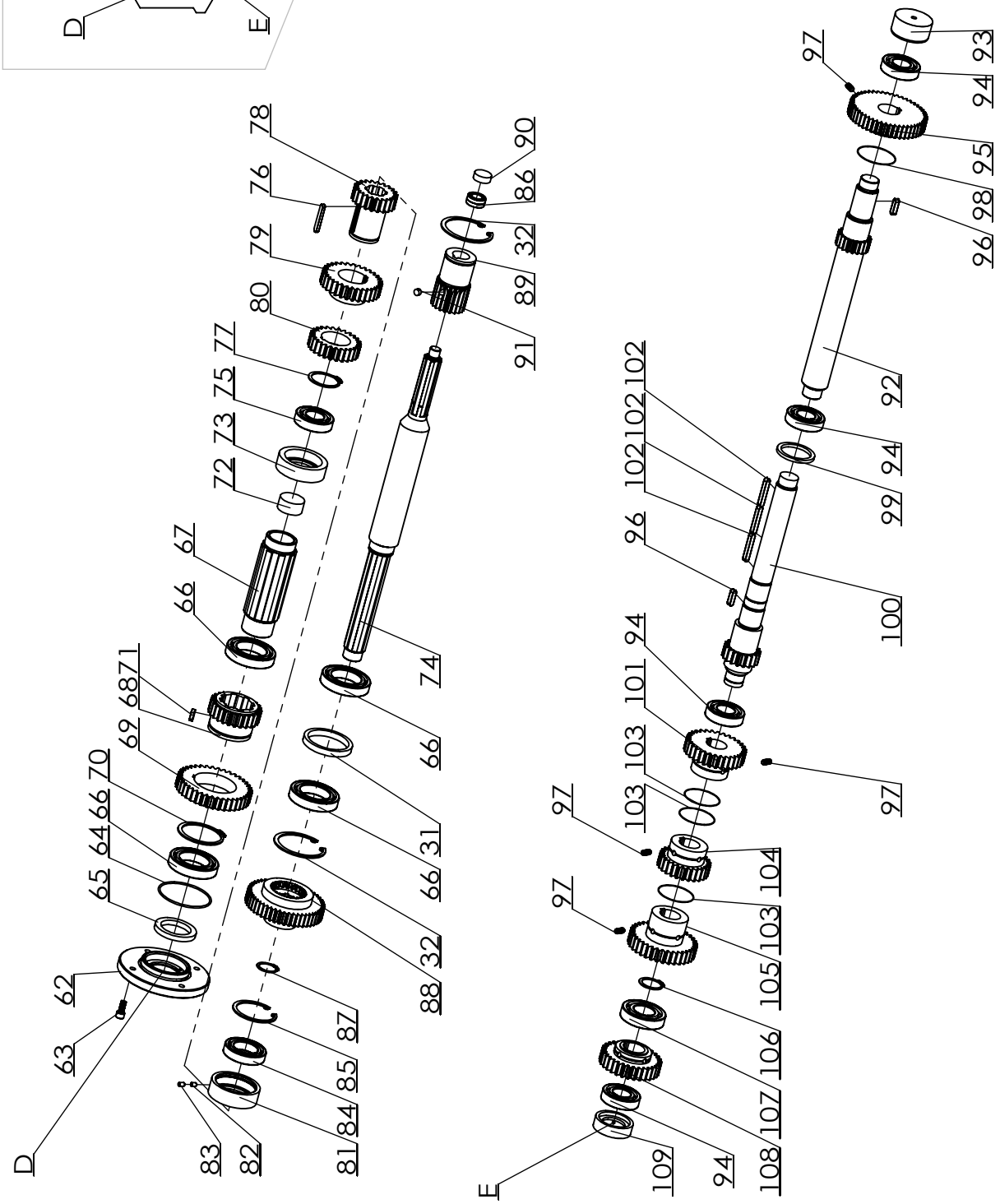
N0.	Name	Qty
04117	Grip	3
04118	Setcrew	3
04119	O-ring	3
04120	Shaft retaining ring	3
04121	Plate	3
04122	Taper pin	3
04123	shifter	1
04124	Bush	1
04125	Bush	1
04126	Shaft	1
04127	Shaft	1
04128	Shifter	1
04129	Shifter	1
04130	Shaft	1
04131	Bush	1
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04133		
04134		

05 Ram Part


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05 Ram Part


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05 Ram part

N0.	Name	Qty
05001	Plate	1
05002	Pressure block	1
05003	Hex. socket head bolt	2
05004	Gib	1
05005	Adjusting screw	1
05006	Gear shaft	1
05007	Bush	1
05008	Setscrew	1
05009	Hex. socket head screw	6
05010	Hex. socket head screw	2
05011	Taper pin	2
05012	Gear box	1
05013	Aluminium plate	1
05014	Connecting plate	1
05015	Plate	1
05016	Washer	1
05017	Plate	1
05018	Oil plug	1
05019	Setcrew	2
05020	Hex. socket head screw	6
05021	Oil plug	1
05022	Motor	1
05023	Washer	4
05024	Hex. socket head screw	4
05025	Hanger	1
05026	Point	1
05027	Rivet	8
05028	Pressure block	1
05029	Washer	4

05 Ram part

N0.	Name	Qty
05030	Bolt	1
05031	Nut	2
05032	Washer	2
05033	Cover	1
05034	Sleeve	1
05035	Lock nut	1
05036	Lock washer	1
05037	Taper pin	3
05038	Oil gauge	1
05039	Oil plug	1
05040	Oil plug	1
05041	Caddice	1
05042	Shaft	1
05043	Rocker	2
05044	Shifter	2
05045	Lever	3
05046	Handle boss	3
05047	Indicator plate	3
05048	Taper pin	3
05049	Shaft retaining ring	3
05050	Sleeve	1
05051	Steel ball	3
05052	Spring	3
05053	Setscrew	3
05054	Shaft	1
05055	handle	3
05056	O-ring	3
05057	Screw	6
05058	Shifter	1

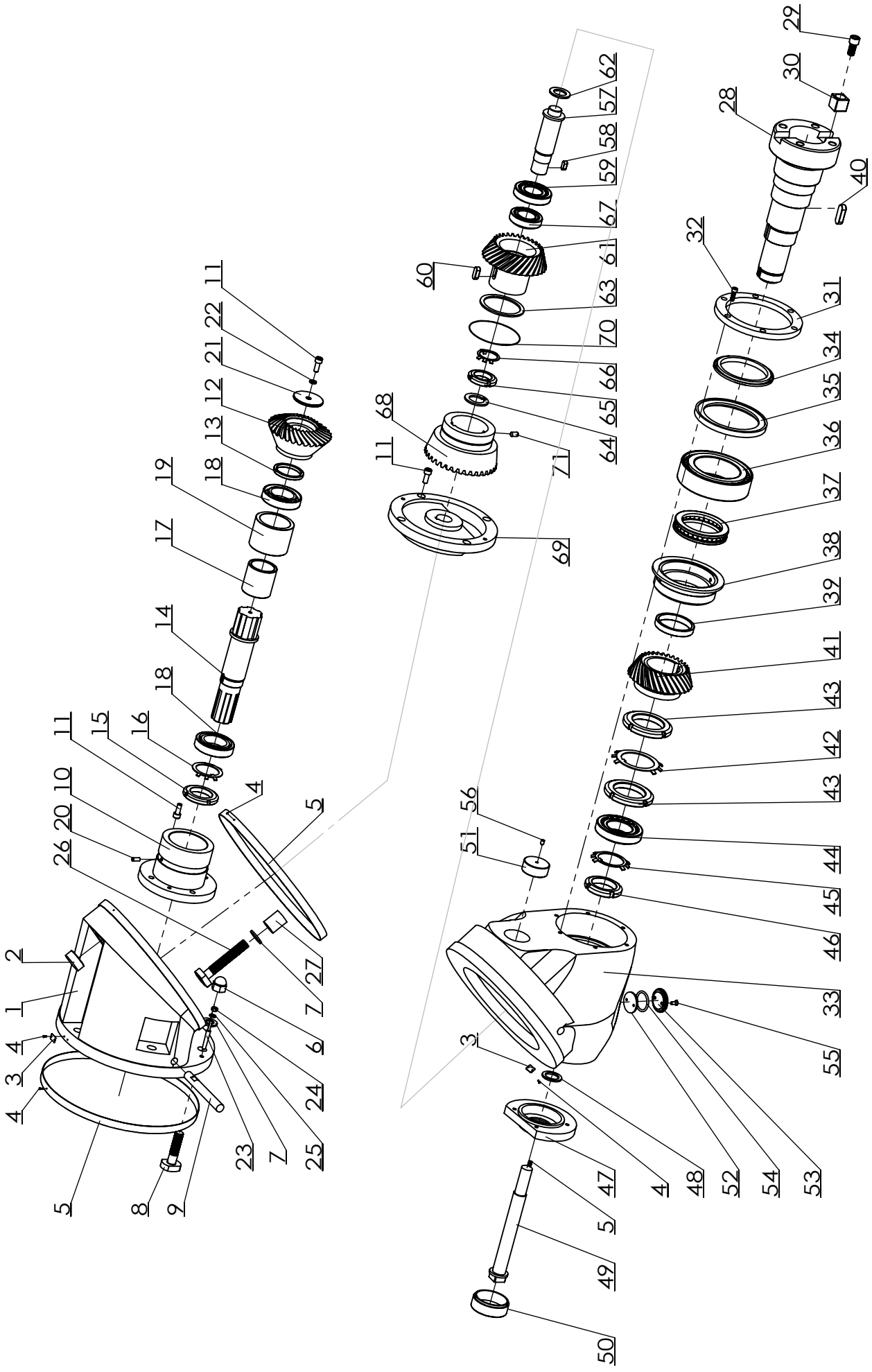
05 Ram part

N0.	Name	Qty
05059	Sleeve	1
05060	Shaft	1
05061	Sleeve	1
05062	Cover	1
05063	Hex. socket head screw	4
05064	O-ring	1
05065	Oil seal	1
05066	bearing	4
05067	Spline sleeve	1
05068	Gear	1
05069	Gear	1
05070	Shaft retaining ring	1
05071	Key	1
05072	Oil plug	1
05073	Bearing house	1
05074	Spline shaft	1
05075	Bearing	1
05076	Key	1
05077	Shaft retaining ring	1
05078	Gear	1
05079	Gear	1
05080	Gear	1
05081	Bush	1
05082	Setscrew	1
05083	Setscrew	1
05084	Bearing	1
05085	Retaining ring for hole	1
05086	bearing	1
05087	Shaft retaining ring	1

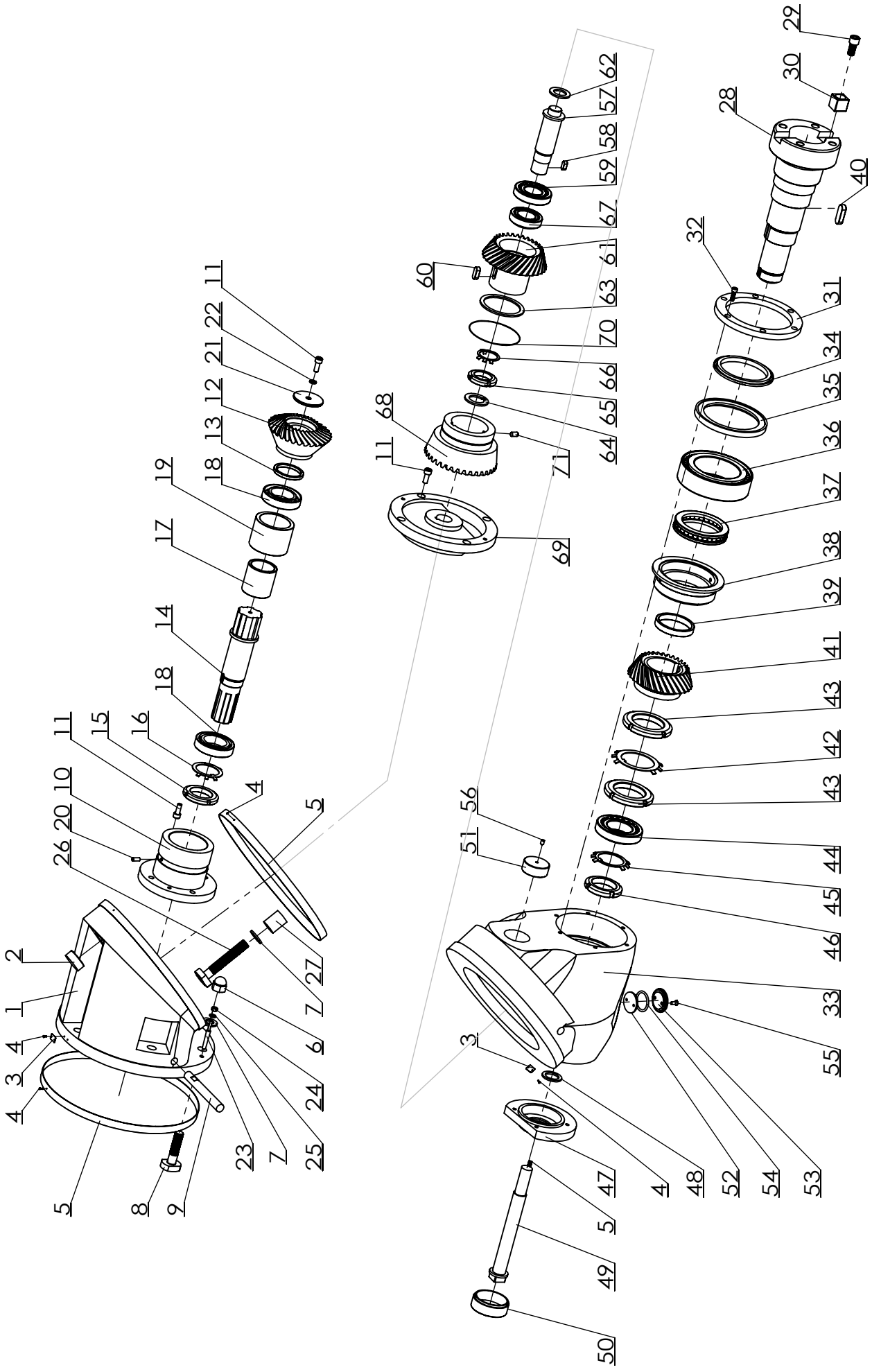
05 Ram part

N0.	Name	Qty
05088	Gear	1
05089	Gear shaft	1
05090	Oil plug	1
05091	Oil plug	1
05092	Gear shaft	1
05093	Plug	1
05094	Bearing	4
05095	Gear	1
05096	Key	2
05097	Setscrew	4
05098	Iron wire	1
05099	Washer	1
05100	Gear shaft	1
05101	Gear	1
05102	Key	3
05103	Iron wire	3
05104	Gear	1
05105	Gear	1
05106	Shaft retaining ring	1
05107	Bearing	1
05108	Gear	1
05109	plug	1
05110		1
05111		1
05112		1
05113		1
05114		1
05115		1
05116		1

06 Headstock Part



06 Headstock Part



06 milling head part

No.	Name	Qty
06001	Rotator	1
06002	Plug	1
06003	Indicator plate	2
06004	Rivet	12
06005	Angle ruler	2
06006	Nut	7
06007	Washer	5
06008	Bolt for T-slot	2
06009	Lever	2
06010	Base	1
06011	Screw	9
06012	Helix gear	1
06013	Washer	1
06014	Spline shaft	1
06015	Nut	1
06016	Lock washer	1
06017	Bush	1
06018	Bearing	2
06019	Bush	1
06020	Taper pin	2
06021	Washer	1
06022	Washer	1
06023	Taper pin	2
06024	Nut	2
06025	Washer	2
06026	Bolt for T-slot	2
06027	Nut	2
06028	Spindle	1
06029	Screw	2

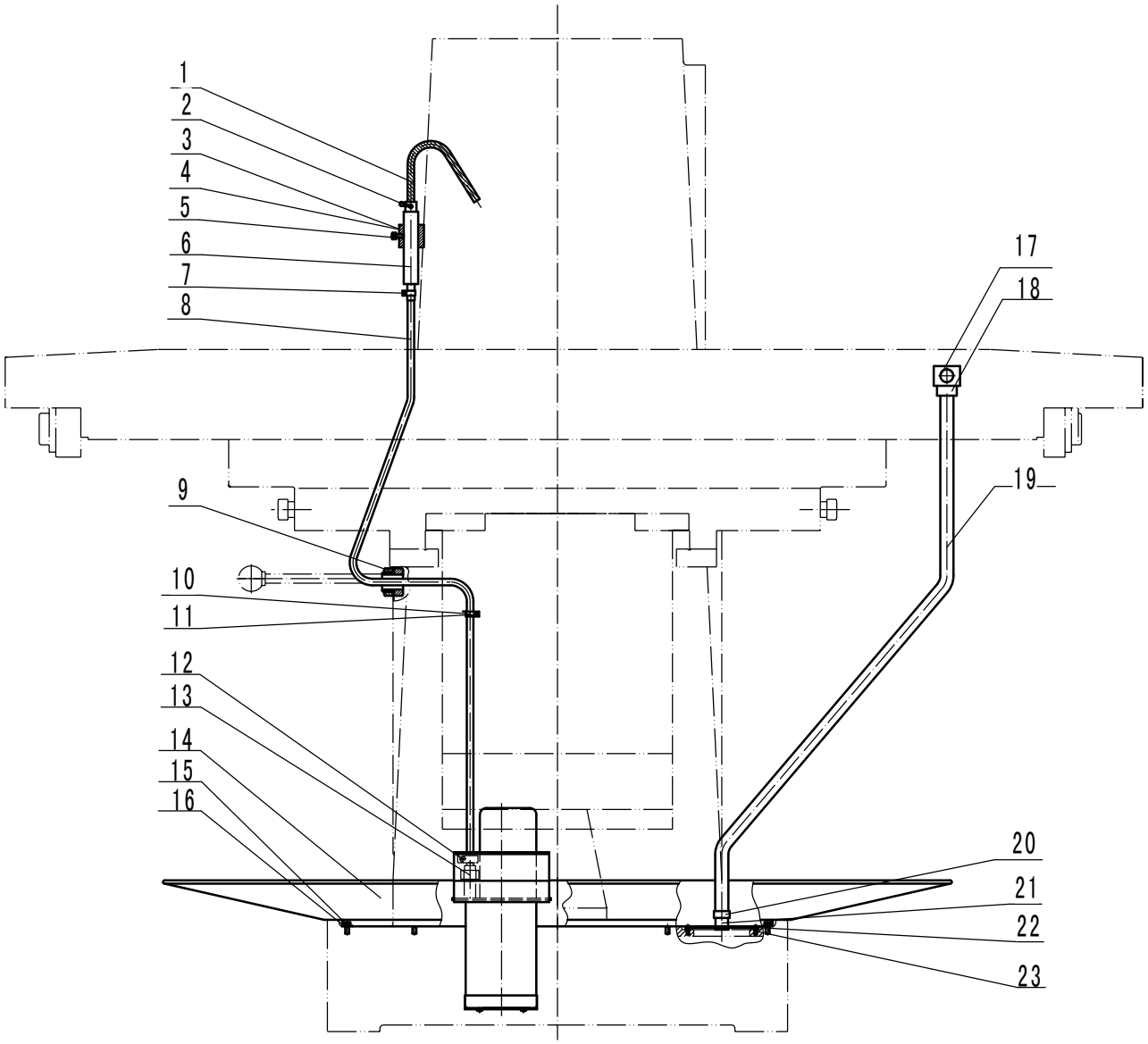
06 milling head part

No.	Name	Qty
06030	Key	2
06031	Cover	1
06032	Screw	6
06033	Headstock	1
06034	Collar	1
06035	Collar	1
06036	Bearing	1
06037	Bearing	1
06038	Bearing sleeve	1
06039	Collar	1
06040	Key	1
06041	Helix gear	1
06042	Lock washer	1
06043	Nut	2
06044	Bearing	1
06045	Lock washer	1
06046	Nut	1
06047	Cover	1
06048	Washer	2
06049	Drawing bolt	1
06050	Cover	1
06051	Plug	1
06052	Cover	1
06053	Cover	1
06054	O-ring	1
06055	screw	2
06056	Setscrew	1
06057	Shaft	1
06058	Key	1

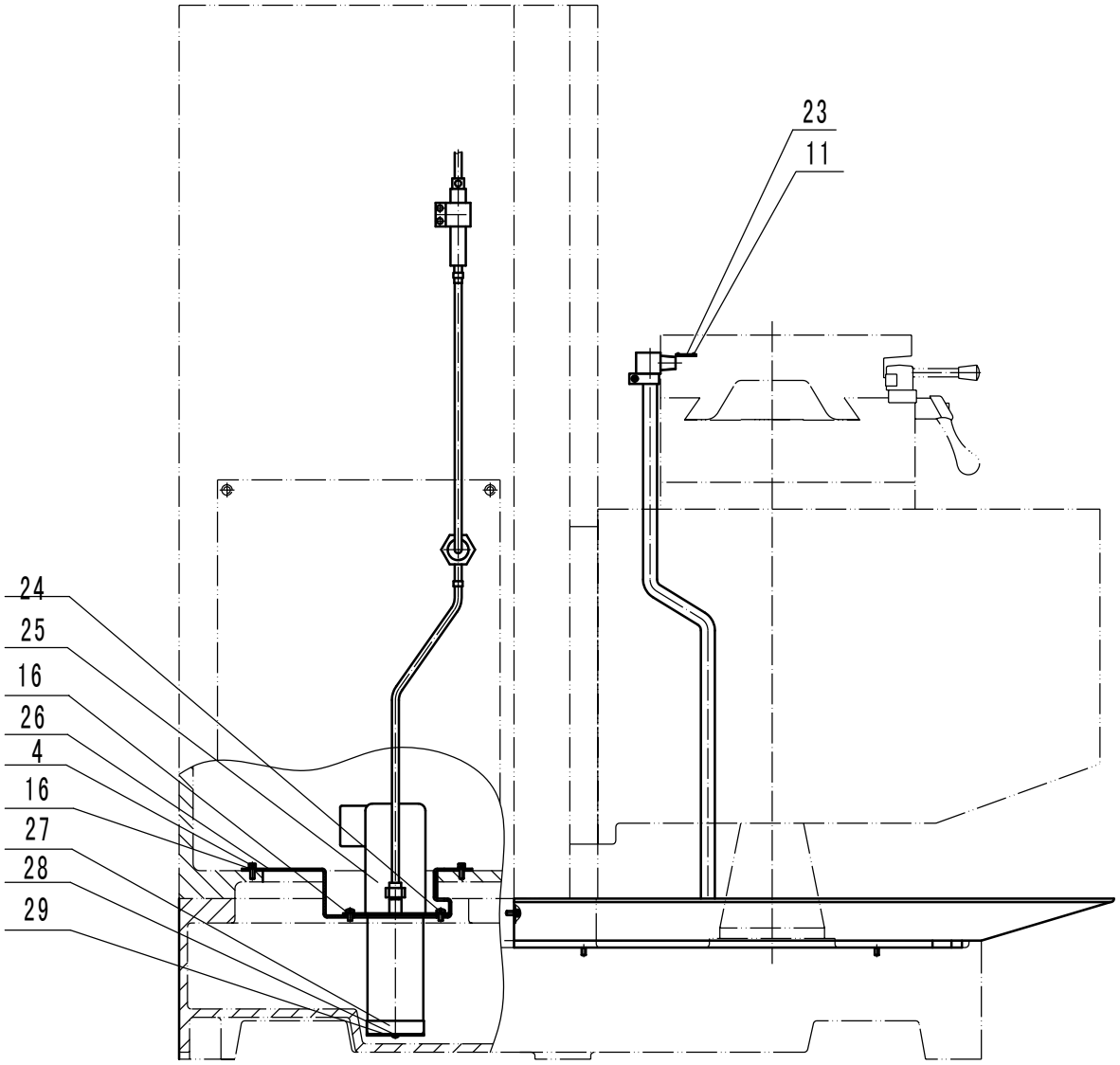
06 milling head part

No.	Name	Qty
06059	Bearing	1
06060	Key	1
06061	Helix gear	1
06062	Washer	1
06063	Washer	1
06064	Washer	1
06065	Nut	1
06066	Lock washer	1
06067	Bearing	1
06068	Helix gear	1
06069	Connecting seat	1
06070	Iron wire	1
06071	Setscrew	1
06072		
06073		
06074		
06075		
06076		
06077		
06078		
06079		
06080		
06081		
06082		
06083		
06084		
06085		
06086		
06087		

09 Cooling parts



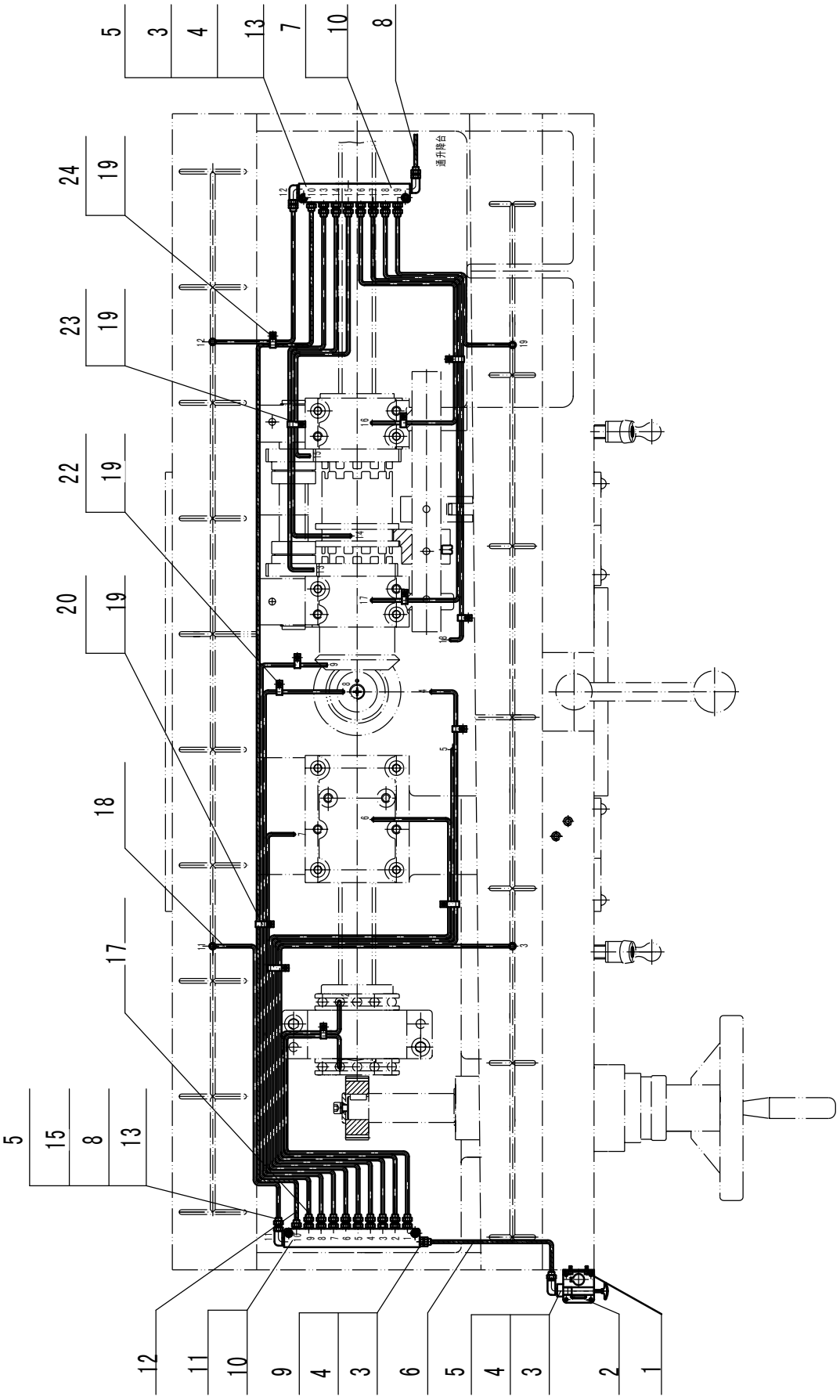
09 Cooling parts



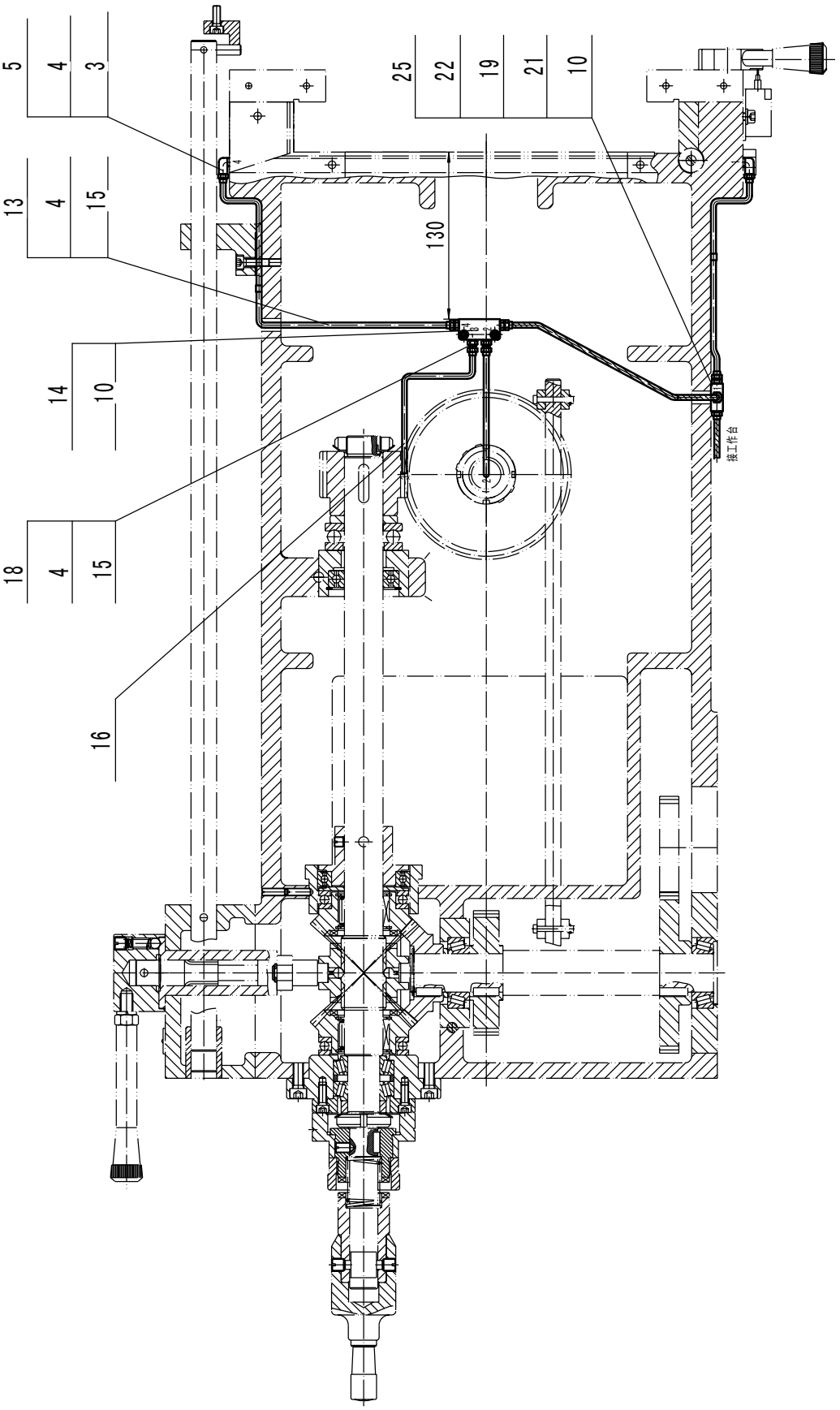
09 10 Cooling & Lubricating parts

No.	Name	Qty
1	Cooling pipe	1
2	Valve	1
3	Fixed boss	1
4	Connecting pipe	1
5	Screw	4
6	Screw	1
7	Hoop	1
8	Hose	1
9	Joint	1
10	Clip	3
11	Screw	5
12	Hoop	1
13	Joint	1
14	Chip tray	1
15	Screw	8
16	Washer	13
17	Joint	1
18	Hoop	2
19	Hose	1
20	Joint	1
21	Plate	1
22	Screw	2
23	Plate	1
24	Screw	4
25	Cooling motor	1
26	Cooling motor base	1
27	Cover	1
28	Net	1
29	Screw	3

10 Lubricating parts



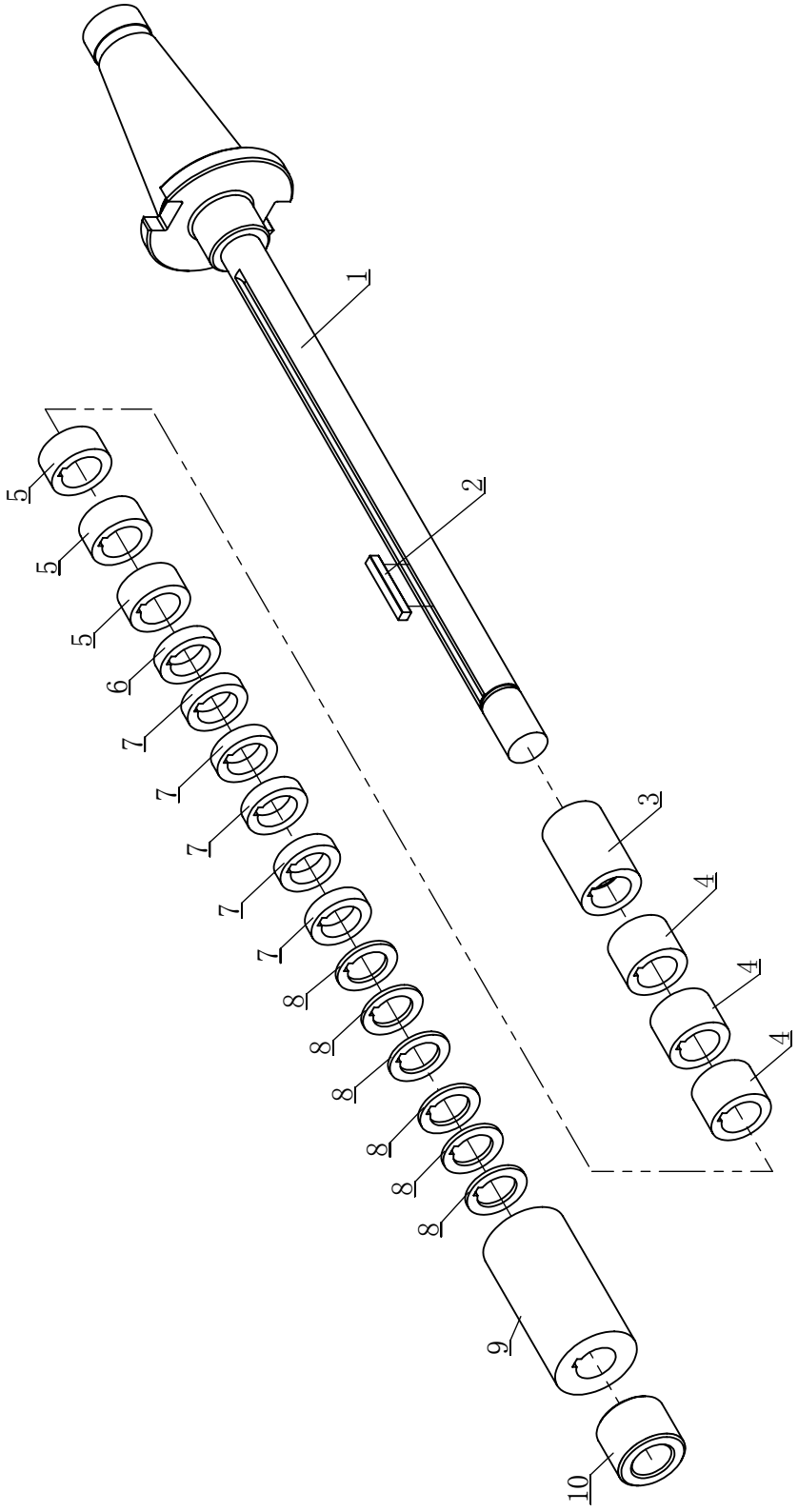
10 Lubricating parts



10 Lubricating parts

No.	Name	Qty
1	Screw	2
2	Manual lubricating motor	1
3	Joint	12
4	Adapter	26
5	Elbow	5
6	Oil pipe	1
7	Distributor	1
8	Oil pipe	1
9	Elbow	1
10	Screw	6
11	Distributor	1
12	Oil pipe	1
13	Gauge	22
14	Distributor	1
15	Nut	14
16	Copper pipe	1
17	Copper pipe	1
18	Nylon pipe	1
19	Screw	14
20	Clipper	4
21	Distributor	1
22	Clip	6
23	Clip	2
24	Clip	2
25	Oil pipe	1
26		1
27		1
28		1
29		3

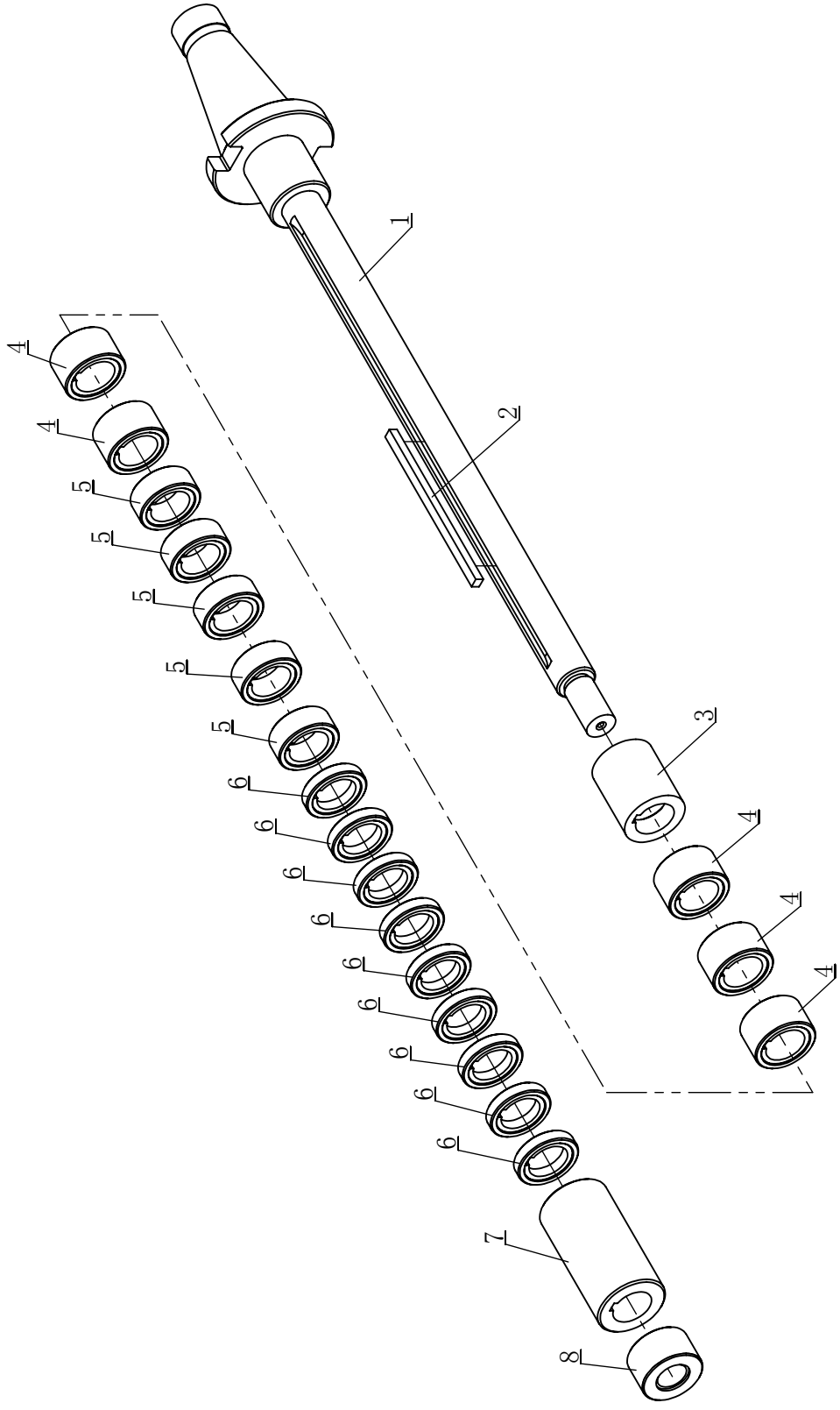
F1 Accessories Part



F1 Accessories part

No.	Name	Qty
F1001	Φ27 Cutter arbor	1
F1002	Key	1
F1003	Distance sleeve	1
F1004	Distance sleeve	3
F1005	Distance sleeve	3
F1006	Distance sleeve	1
F1007	Adjusting washer	5
F1008	Adjusting washer	6
F1009	Sleeve	1
F1010	Nut	1

F1 Accessories Part



Accessories part

No.	Name	Qty
F1001	Φ32 Cutter arbor	1
F1002	Key	1
F1003	Distance sleeve	1
F1004	Distance sleeve	5
F1005	Distance sleeve	5
F1006	Distance sleeve	9
F1007	Sleeve	1
F1008	Nut	1